QuaILLL: Quaternion Ideal LLL and BKZ

Joshua Limbrey, Cong Ling, and Christian Porter

Department of Electronics and Electrical Engineering
Imperial College London
{j.limbrey24,c.ling,c.porter17}@imperial.ac.uk

Abstract. The current state of the art for cryptanalysis generic rank-2 module LIP schemes invokes an SVP oracle on the canonical real embedding, discarding the quaternionic structure made available by the reduction of rank-2 module LIP to the reduced-norm Principal Ideal Problem (nrd-PIP) over quaternion algebras.[1] We address this gap by giving, to our knowledge, the first lattice reduction algorithms over quaternion rings applied in a cryptographic setting, and the first description of quaternion BKZ. We extend the celebrated LLL algorithm to leverage algebraic properties of quaternion orders and novel post-processing steps to design an LLL algorithm for lattices in not-necessarily-maximal orders. The strategy is to reduce over the Euclidean overlattice and then post-process, giving two routines: one returning a basis of a sublattice with the best bounds, the other a true basis of the original lattice at the cost of output quality. We further consider blocksize two BKZ as a generalisation of the LLL algorithm, and then extend this to arbitrary blocksize; utilising results on the shortness of Gauss and HKZ reduced bases and the relationship of successive minima for our specific sublattice. We then apply these algorithms to ideal lattices arising from nrd-PIP, including those instances given by rank-2 MLIP over cyclotomic fields such as Hawk, via a modification of the canonical embedding that preserves both dimension and quaternionic structure. This allows us to reduce a lattice basis of rank a constant factor of four smaller than the standard real embedding, improving basis bounds and asymptotic complexity in the generic setting.

Keywords: Quaternion algebras · LLL reduction · BKZ reduction · Cyclotomic fields · Lattice isomorphism problem

1 Introduction

Since the advent of Shor's algorithm [34], cryptographers have sought hardness assumptions distinct from the classical discrete logarithm and prime factorisation problems. At the forefront of this search stand schemes reliant on the hardness of so-called lattice problems; one might conjecture this is due to the fact

[1] We note, that since writing, this is no longer the case for certain instances, such as Hawk.

they often enjoy worst-case to average-case reductions, and reductions to long-studied and currently intractable problems such as the shortest vector problem (SVP). Efforts to both develop schemes built with hardness deriving from the intractability of lattice problems as well as cryptanalyse such schemes and solve these problems have been further motivated by NIST's call for post quantum schemes for standardisation of both key encapsulation mechanisms (KEMs) and digital signatures.

To understand the security of these schemes, it is vital that both the schemes and the problems they are based on are thoroughly cryptanalysed. In the case of lattices, a core methodology in the toolbox of the cryptanalyst is that of lattice reduction; namely, the use of algorithms such as LLL [20], BKZ [32, 33], and HKZ [18]. These three algorithms lie on a scale of efficiency versus output quality — the celebrated LLL algorithm exists at one end with a polynomial runtime but exponential approximation factor, HKZ the opposite, and BKZ lies between the two depending on blocksize used. Due to its polynomial runtime, LLL is frequently used in practice as a pre-processing step, and an LLL reduced basis will be given as input to the HKZ and BKZ algorithms. The goal of these algorithms is to recover some notion of a 'reduced' basis — typically this will amount to shortness of basis vectors as well as orthogonality of basis vectors with respect to one another.

Despite having been researched fairly extensively, e.g. [29, 15], the Lattice Isomorphism Problem (LIP) did not see cryptographic use until the work of Ducas and Van Woerden [9] gave efficient methods for sampling for explicit constructions of cryptographic schemes. This in turn led to the construction of a signature scheme by the name of Hawk [3] utilising a structured variant of LIP in order to increase efficiency that was submitted to the NIST call for additional signature schemes. Whilst Hawk was the motivation for this work, reaching the third round of the NIST call for additional signatures, it was subsequently withdrawn due to the works [37, 24]; our techniques may be more broadly applied to rank-2 MLIP. We also note that these works were *not* a full break on Hawk or rank-2 MLIP, though the immediate repair renders Hawk uncompetitive in terms of parameters compared to the other NIST submissions.

The Lattice Isomorphism Problem (in its search form) asks one to find the orthogonal mapping between two given lattices $\mathcal{L}_1$ and $\mathcal{L}_2$ if such a mapping exists, and the decision case's to simply output a yes or no for the transformation's existence. Note, that as is the case in most computational lattice problems, the lattices in question are given as basis matrices. This problem has been studied extensively, particularly in the case of so-called 'remarkable' lattices such as the $\mathbb{Z}^n$ lattice [9, 7], as well as more general cases [15]. All such algorithms to solve this problem, however, amount to an attempt to enumerate short vectors in both challenge lattices before comparing one to the other. It can readily be seen from this description of solving LIP, that LIP in practice amounts to solving SVP; returning us to the task of lattice reduction, and the understanding and improvement of its algorithms. Cryptographers may be attracted to LIP due to

its 'simple' nature, as well as its ability to utilise lattices with specially chosen decoding properties.

Rarely in lattice-based cryptography is it practical to use plain lattice problems, and so structure is introduced to the scheme in order to improve the efficiency of algorithms such as key generation, signing, and encryption. Another benefit of this added structure is that it can decrease the key sizes required to achieve a desired level of security, such as in the case of Hawk, where rank two modules over power-of-two cyclotomic fields are used, allowing the authors to utilise techniques already existing in the literature [16, 30, 31].

The use of this structure leads us to question whether the security of the scheme is impacted by its introduction, and whether there exists any notable decrease in security when compared to standard LIP. One notable inroad into understanding this problem was presented in the line of work [5, 10, 6] (which we will now reference by the latter, a merging of the first two), which gives a reduction from the rank-2 module LIP problem to a problem of solving the norm reduced principal ideal problem (nrd-PIP) in quaternion algebras.

PIP has received a large amount of scrutiny as a computational number theoretic problem, with work by Gentry and Szydlo developing a polynomial time algorithm to solve a variant of PIP over the ring $\mathbb{Z}[\zeta_p]$ for prime p and attack an NTRU based scheme [14]. This was then followed by a string of works by Lenstra and Silverberg [21, 22, 17], in which the Gentry-Szydlo algorithm is used to solve LIP over cyclotomic fields $\mathbb{Q}(\zeta_m)$ for $m \in \mathbb{N}$, and further generalised to CM orders. This relies on commutativity, however, and so is non-trivial to apply in the case of a non-commutative algebra such as the quaternions; exactly the setting that arises from Hawk.

Because of this, it was the case that breaking Hawk amounted to an SVP oracle. This oracle was called against the ideal considered as a lattice under an embedding into $\mathbb{R}^{4n}$, where n is the degree of the maximal totally real subfield of the underlying power-of-two cyclotomic field. Because of this embedding, no structure was able to be leveraged from the ideal or quaternion structure, all of which is lost in the embedding process. Prior to July 2026, the best such attack, due to Ducas [7], required exact SVP calls in dimension $2n + 1$. The works [37, 24] reduced this to exact SVP in dimension $n + 1$ and $3n/2 + 1$ respectively in the special case of Hawk, utilising the power-of-two cyclotomic structure present. Our work addresses the gap that therefore arises between these dimension-reducing attacks for the power-of-two case and the general case; and is able to run against rank-2 MLIP instances over cyclotomics with conductors m such that $3|m$ or $4|m$. Our algorithms instead reduce the ideal directly as a rank-n quaternion lattice, rather than a lattice in $\mathbb{R}^{4n}$, retaining its quaternionic structure; we are not able to solve exact-SVP in this lattice, however, and so our approach currently *does not* directly impact the security of these schemes.

1.1 Our Contribution

We begin by extending the quaternionic generalisation of LLL to lattices as first given by [26], and the bounds and complexity further studied in [35] to provide

reduced bases for lattices over natural orders of rational quaternion algebras. We do this by considering this lattice as a sublattice contained within the lattice over the maximal and Euclidean order that contains the natural order, in which we are able to utilise the quaternion LLL already in literature and then provide a number of post-processing steps to recover a solution in the lattice we are interested in. These recovery steps differ depending on the end goal: one of these post-processing steps will return a basis of some further sublattice but has the best bound for the shortest basis vector, and the other a true basis of this lattice at the cost of output quality — this may be attractive if further reduction is to take place. Additionally, viewing LLL as a specific case of BKZ with blocksize two, we prove optimality of Gauss reduction for Euclidean quaternion lattices, and use this to define a blocksize two BKZ algorithm, which can be thought of as a sister algorithm to LLL. This result is then extended from Gauss reduction to HKZ, and thereby arbitrary blocksize BKZ.

We then study solving the quaternion ideal SVP instances arising from the nrd-PIP problem in quaternion algebras utilising the algebraic structure present instead of embedding into a real lattice. We consider a rank-2 module LIP instance over a cyclotomic field and using the work of [6] to give an embedding into a lattice in a Euclidean quaternion order when the conductor of the cyclotomic field is divisible by 3 or 4. This embedding preserves the structure of the quaternion algebra present in the nrd-PIP instance, and gives a quaternion ideal lattice.

By retaining this quaternion structure, we are able to take advantage of the reduced dimension by a constant factor of four when compared to the real embedding, improving our shortest basis vector bound and orthogonality defect. This is also notable as the dimension is the critical factor when considering the algorithmic complexity of LLL, whose runtime is polynomial in dimension.

1.2 Related Work

The literature has focused primarily on tailoring standard reduction algorithms to the module setting, and despite the use of quaternion lattice reduction being used in coding theory [35, 36], it has not yet seen use in a cryptographic setting. We do note that there have been attempts to use very closely related symplectic structure arising in NTRU lattices [12, 2].

There exists other work that studies using the module structure present in schemes such as Hawk for lattice reduction and how this affects their security [19, 8]. These works conclude that the efficacy of cyclotomic module lattice reduction depends heavily on the conductor of the cyclotomic field. We note rank-2 Module LIP using power-of-two cyclotomics (as used in Hawk), the results of [8] suggest that there is little advantage to leveraging module structure available, [28] agree but show that the speed of this convergence may be faster than traditional methods. We note that for other conductors, such as powers of three, five, or composite powers, module BKZ may provide an advantage.

Further cryptanalytic efforts of rank-2 module LIP in general were given, with [25, 1] giving polynomial time algorithms to solve rank-two Module LIP

in number fields that are not totally complex. The current state of these two lines of research encourage us to look to the alternative quaternionic structure presented by PIP instead of the more complex module structure in the Hawk.

Finally, work published by Anthropic [37] detailed a key recovery attack on Hawk; they first detail a non-trivial automorphism stemming from the Galois involution $\zeta \mapsto -\zeta$. Such an automorphism is computable from a public rank-$2n$ lattice isometric, up to scaling, to $\mathbb{Z}^{n+1} \oplus \sqrt{2}\mathbb{Z}^{n-1}$, from which it may be recovered by Ducas' reduction for near-hypercubic lattices [7] with polynomially many exact-SVP calls in dimension $n+1$. This non-trivial automorphism may then be used with [13] to recover the key in polynomial time. This work was closely followed by [24], taking a similar approach. They instead start from the nrd-PIP instance arising from [6], from which they consider the trace-zero subspace; conjugation by the secret generator fixes the totally real subfield pointwise, and so acts as an isometry upon this subspace. This yields a rank-3 module LIP instance over this subfield, whose underlying lattice in the case of Hawk is isometric to $\mathbb{Z}^{2n+1} \perp \sqrt{2}\mathbb{Z}^{n-1}$; again, the same technique allows the secret to be recovered with polynomially many exact-SVP calls in dimension at most $3n/2+1$. Following these attacks, Hawk was withdrawn from the NIST standardisation process.

As already mentioned, PIP has seen plenty of scrutiny in the general case through the Gentry-Szydlo algorithm, as well as work on ideal SVP [27, 11] but little work that is applicable to the quaternion setting. One such example, [23], addresses understanding easy cases of the quaternion ideal SVP problem as arise from nrd-PIP.

2 Preliminaries

Throughout, we will use bold upper case to denote matrices, and bold lower case to denote vectors, and lower case letters to denote scalar values. We will use subscript to denote the i-th column, or i,j-th entry of a matrix.

2.1 Lattices

Definition 1 (Lattice). *A lattice is an additive discrete subgroup of $\mathbb{R}^n$ for some $n \in \mathbb{N}$. A lattice is generated by a basis $\mathbf{B} = \{\mathbf{b}_1, ..., \mathbf{b}_k\}$, a set of linearly independent vectors with $k \leq n$; if $k = n$, then we say the lattice is full rank, and we denote the lattice generated by $\mathbf{B}$ by $\mathcal{L}(\mathbf{B}) = \{\sum_{i=1}^{k} z_i \mathbf{b_i} \mid z_i \in \mathbb{Z}\}$.*

Definition 2 (R-Lattice). *Let R be a domain with field of fractions F. An R-lattice in an F-vector space V is a finitely generated R-submodule that spans V over F.*

An R-lattice may still be represented by a basis $\mathbf{B} \in K^{n \times k}$ in the same way a lattice may be if R is a PID. In this case, $\mathcal{L}(\mathbf{B}) = \{\sum_{i=1}^{k} r_i \mathbf{b}_i \mid r_i \in R\}$.

Remark 1. We may shorten 'R-lattice' to 'lattice' when it is clear from context the ring we are working over.

Definition 3 (Successive Minima). *For an R-lattice $\mathcal{L}$, the i-th successive minima, $\lambda_i(\mathcal{L}) := \inf\{r \mid \dim\{\text{span}_F\{\mathcal{L} \cap \mathcal{B}_r\}\} = i\}$, where $\mathcal{B}_r$ is a ball of radius r, and dim$\{\cdot\}$ denoting the number of linearly independent vectors in the set, or the dimension of the vector space.*

In other words, the i-th successive minima is the smallest radius r such that i linearly independent vectors can fit inside a ball of radius r.

Definition 4 (Shortest Vector Problem (SVP)). *Given a lattice $\mathcal{L}$, find a vector $\mathbf{v} \in \mathcal{L}$ such that $\mathbf{v}$ is of length $\lambda_1(\mathcal{L})$.*

Definition 5 (Hermite SVP). *Let $\mathcal{L}$ be a rank n lattice and $\gamma \geq 1$. The γ-Hermite SVP problem asks to find a non-zero vector $\mathbf{v} \in \mathcal{L}$ such that*

$$\|\mathbf{v}\| \leq \gamma \cdot \text{vol}(\mathcal{L})^{1/n}.$$

2.2 Number Fields

An algebraic number field F is a finite extension of the field of rational numbers $\mathbb{Q}$, and is isomorphic to $\mathbb{Q}[X]/P(X)$ for an irreducible monic polynomial $P(X)$ of $\mathbb{Q}[X]$. We say that F has degree $d = [F : \mathbb{Q}] = \deg(P(X))$.

A number field of degree d has d embeddings from $F \to \mathbb{C}$, and is comprised of r real embeddings and c complex embeddings, where $r + 2c = d$. If an embedding $\sigma_i : F \to \mathbb{R}$, then it is a real embedding, and is complex otherwise. Any complex embedding σ_j can be composed with complex conjugation to obtain a different complex conjugation $\overline{\sigma_j} = \sigma_{j'}$. We call a number field totally real if all its embeddings are real, and totally complex if none are real.

The canonical embedding of a number field F is defined as

$$\sigma(f) := (\sigma_1(f), ..., \sigma_d(f)) \in \mathbb{C}^d, \quad \forall f \in F.$$

We will primarily concern ourselves with the cyclotomic fields, which are obtained by adjoining a primitive m-th root of unity, $L = \mathbb{Q}(\zeta_m)$. We note that these have degree $[\mathbb{Q}(\zeta_m) : \mathbb{Q}] = \varphi(m)$, and their ring of integers is $\mathbb{Z}[\zeta_m]$. Further, we note that the cyclotomic field L is totally complex, and $K = \mathbb{Q}(\zeta_m + \zeta_m^{-1})$ its maximal totally real subfield.

2.3 Quaternions

Definition 6 (Quaternion Algebra). *Let K be a number field of degree $n = [K : \mathbb{Q}]$, and let $a, b \in K$. Then $\mathcal{A} = \left(\frac{a,b}{K}\right)$ denotes the quaternion algebra generated by i, j with relations $i^2 = a$, $j^2 = b$, and $ij = -ji = k$. An element $q \in \mathcal{A}$ may be written $q = q_1 + q_2 i + q_3 j + q_4 k$.*

If a, b are both totally negative in K and K totally real, then $\mathcal{A}$ is called totally definite. Henceforth, all quaternion algebras considered will be totally definite.

Definition 7 (Order). *An order $\mathcal{O} \subset \mathcal{A}$ is an $\mathcal{O}_K$-lattice that is also a ring. We say that an order is maximal if it is not properly contained in another order.*

Definition 8 (Left/Right Ideal). *Let $\mathcal{O} \subset \mathcal{A}$ be an order in a quaternion algebra. A left (right resp.) ideal of $\mathcal{O}$ is an additive subgroup $\mathcal{I} \subset \mathcal{A}$ such that $\mathcal{O}\mathcal{I} \subset \mathcal{I}$ ($\mathcal{I}\mathcal{O} \subset \mathcal{I}$ resp.). A left (right resp.) ideal is principal if it is generated by a single element $\alpha \in \mathcal{A}$, $\mathcal{I} = \mathcal{O}\alpha$ ($\mathcal{I} = \alpha\mathcal{O}$ resp.).*

It is important to note that quaternions (in general) do not commute under multiplication, and as such, do not form a field, but a skew-field.

We will use the following notation conventions given a quaternion element $q = q_1 + q_2 i + q_3 j + q_4 k \in \mathcal{A} = (\frac{a,b}{K})$:

- The canonical involution, conjugation by $\overline{q} = q_1 - q_2 i - q_3 j - q_4 k$
- The reduced norm by $\mathrm{nrd}_{\mathcal{A}/K}\, q = q\overline{q} = \overline{q}q = q_1^2 - aq_2^2 - bq_3^2 + abq_4^2$

Since the totally definite case we are working in provides us with a division algebra [38], division can be done using multiplication with the inverse element, $q^{-1} = \overline{q}(\overline{q}q)^{-1}$. Addition of quaternion elements is performed component-wise; multiplication follows from the relations of i, j, k extended by K-linearity.

When working with quaternion valued matrices and vectors, we will use the Hermitian transpose, $\mathbf{v}^H = (\overline{\mathbf{v}})^T$; and define

$$\|\mathbf{v}\|_{\mathcal{A}/K}^2 = \mathbf{v}^H\mathbf{v} = \sum \mathrm{nrd}_{\mathcal{A}/K}(\mathbf{v}_i) \in K,$$

and

$$\langle \mathbf{v}, \mathbf{w} \rangle_{\mathcal{A}} = \mathbf{w}^H\mathbf{v} \in \mathcal{A}.$$

Definition 9 (Quaternion Lattice). *Let $\mathcal{A}$ be a quaternion algebra over K, and let $\mathcal{O} \subset \mathcal{A}$ be a quaternion order. A quaternion lattice is an $\mathcal{O}$-lattice in $\mathcal{A}^n$, i.e. a finitely generated $\mathcal{O}$-submodule of $\mathcal{A}^n$ that spans $\mathcal{A}^n$ over K.*

When $\mathcal{O}$ is a principal ideal domain, every quaternion lattice is free as an $\mathcal{O}$-module and admits a basis matrix $\mathbf{B} \in \mathcal{A}^{n\times k}$, in which case we may write:

$$\mathcal{L}(\mathbf{B}) = \{\sum_{i=1}^{k} q_i \mathbf{b}_i \mid q_i \in \mathcal{O}\}.$$

A quaternion ideal of $\mathcal{O}$ is a quaternion $\mathcal{O}$-lattice in $\mathcal{A}$ with rank $n = 1$.

We will consider a subset of these quaternion ideal lattices that naturally arise from the rank-2 Module-LIP problem as described in [6]. We will later describe how such ideals arise, and how one can efficiently construct a quaternion-ideal lattice from this whilst maintaining quaternionic structure.

2.4 Euclidean Quaternion Lattices

Let us denote by $N : \mathcal{A} \to \mathbb{Q}$ the reduced norm map $N = N_{K/\mathbb{Q}} \circ \mathrm{nrd}_{\mathcal{A}/K}$.

Definition 10 (Norm-Euclidean Orders [4]). *An order Λ is norm-Euclidean if for any $a, b \in \Lambda$ and $b \neq 0$, there exists some $q \in \Lambda$ such that*

$$N(a - bq) < N(b)$$

The reader may notice that we have not talked about left/right Euclidean in the above definition despite working in a non-commutative algebra. It is the case that if an order is right-Euclidean then it is necessarily also left-Euclidean. Further, if it is Euclidean, it is maximal, and all maximal orders of the algebra are Euclidean [4].

Remark 2. We will use Λ to denote a maximal and Euclidean order of a quaternion algebra.

Definition 11. *Let Λ be an order of $\mathcal{A}$. For any $\xi \in \mathcal{A}$, we set*

$$m_\Lambda(\xi) = \inf_{\lambda \in \Lambda} N(\xi - \lambda),$$

and we call it the local Euclidean minimum of Λ at ξ. We define the Euclidean minimum of Λ by

$$\mathcal{M}(\Lambda) = \sup_{\xi \in \mathcal{A}} m_\Lambda(\xi).$$

We call Q_Λ a quantisation function for Λ if $N(\xi - Q_\Lambda(\xi)) = m_\Lambda(\xi)$ for all $\xi \in \mathcal{A}$.

Thus, given a quaternion $q \in \mathcal{A}$ and an order Λ of $\mathcal{A}$, the quantisation function is

$$Q_\Lambda : \mathcal{A} \longrightarrow \Lambda, \qquad Q_\Lambda(q) = \arg\min_{h \in \Lambda} N(q - h).$$

Proposition 1 ([4]). *An order Λ is Euclidean if and only if $\mathcal{M}(\Lambda) < 1$.*

Lemma 1 ([4]). *There exist, up to equivalence, exactly three norm Euclidean orders in totally definite quaternion algebras over $\mathbb{Q}$. Namely, these are*

$$\mathcal{A}_1 = \left(\frac{-1,-1}{\mathbb{Q}}\right) \text{ with order } \Lambda_1 = \mathbb{Z}\left[i, j, \frac{1+i+j+k}{2}\right],$$
$$\mathcal{A}_2 = \left(\frac{-1,-3}{\mathbb{Q}}\right) \text{ with order } \Lambda_2 = \mathbb{Z}\left[i, \frac{i+j}{2}, \frac{1+k}{2}\right],$$
$$\mathcal{A}_3 = \left(\frac{-2,-5}{\mathbb{Q}}\right) \text{ with order } \Lambda_3 = \mathbb{Z}\left[\frac{1+i+j}{2}, j, \frac{2+i+k}{4}\right]$$

Notably, these orders are also all maximal.

The Euclidean minima, discriminants, indices, and quantisation functions for these orders are given in Appendix A.

The other type of order we will consider is the natural order of an algebra

$$\mathcal{N} := \mathcal{O}_K + i\mathcal{O}_K + j\mathcal{O}_K + k\mathcal{O}_K \subset \Lambda \subset \mathcal{A}.$$

For rational quaternion algebras, these are simply integer valued quaternions. For the rest of the paper, we will use $\mathcal{N}_\iota \subset \Lambda_\iota \subset \mathcal{A}_\iota, \iota = 1, 2$ to refer to the natural order, maximal order, and rational quaternion algebras respectively; $\mathcal{O} \subset \mathcal{A}$ will denote an arbitrary order.

2.5 Lattice Reduction

Throughout, we will use the following notation for any set of vectors $\mathbf{b}_1, ..., \mathbf{b}_n \in \mathcal{A}^m$:

$$\mathbf{b}_i(j) := \mathbf{b}_i - \sum_{k=1}^{j-1} \mu_{i,k}\mathbf{b}_k(k), \qquad \mathbf{b}_1(1) := \mathbf{b}_1,$$
$$\mu_{i,k} := \frac{\langle \mathbf{b}_i, \mathbf{b}_k(k)\rangle_{\mathcal{A}}}{\|\mathbf{b}_k(k)\|^2_{\mathcal{A}/K}} \in \mathcal{A},$$

for any $1 \le j \le i \le n$, $1 \le k < i \le n$, to refer to our Gram-Schmidt vectors and coefficients respectively. Given a basis $\mathbf{B} \in \mathcal{A}^{n\times m}$, the set of vectors $\mathbf{b}_1(1), ..., \mathbf{b}_n(n) \in \mathcal{A}^m$ is called the Gram-Schmidt Orthogonalisation (GSO).

Definition 12 (Lattice Volume). *A quaternion lattice $\mathcal{L}(\mathbf{B})$ defined over an order $\mathcal{O}$ has volume*

$$\begin{aligned}\mathrm{vol}(\mathcal{L}) &= \sqrt{N_{K/\mathbb{Q}}(\det(\mathbf{B}^H\mathbf{B})) \cdot |\operatorname{disc}(\mathcal{O}/\mathbb{Z})|} \\ &= \sqrt{N_{K/\mathbb{Q}}(\prod_{i=1}^{n} \|\mathbf{b}_i(i)\|^2_{\mathcal{A}/K}) \cdot |\operatorname{disc}(\mathcal{O}/\mathbb{Z})|}.\end{aligned}$$

Definition 13 (Orthogonality Defect of a Lattice Basis). *The orthogonality defect of a lattice basis measures relative orthogonality of lattice vectors and for an $\mathcal{O}$-lattice $\mathcal{L}$ with basis $\mathbf{B}$ is defined by*

$$\Omega_{\mathcal{L}}(\mathbf{B}) := \frac{\sqrt{N_{K/\mathbb{Q}}(\prod_{i=1}^{n} \|\mathbf{b}_i\|^2_{\mathcal{A}/K})}}{\mathrm{vol}(\mathcal{L})}.$$

When $K = \mathbb{Q}$, $\mathcal{A} = \mathbb{Q}$, and $\mathcal{O} = \mathbb{Z}$, the above definitions recover the standard ones for $\mathbb{Z}$-lattices.

Definition 14. *Let $\mathbf{B}$ be a basis for the rank n lattice $\mathcal{L}$. We define the linear map π_i as the projection away from the span of $\mathbf{B}_{1:i-1}$. Then the projected sublattice $\pi_i(\mathcal{L}_{[i:j]})$ is the lattice with the basis $\pi_i(\mathbf{B}_{[i:j]}) = (\pi_i(\mathbf{b}_i), ..., \pi_i(\mathbf{b}_j))$.*

3 LLL Reduction over Euclidean Quaternion Orders

As described in [26], the quantisation functions arising in Euclidean quaternion algebras give a notion of 'rounding', allowing the size reduction step (Algorithm 2), and can be directly applied to give LLL reduction over Euclidean quaternion algebras, described in Algorithm 1.

3.1 Quaternion LLL

Definition 15 (Quaternion LLL-Reduction over Euclidean Orders). *Let $\mathcal{A}$ be a totally definite quaternion algebra over a number field K, and $\mathbf{B}$ the basis for the lattice $\mathcal{L}$ over the Euclidean order $\Lambda \subset \mathcal{A}$.*

We say that $\mathbf{B}$ is quaternion-LLL-reduced over Λ if the following two conditions are satisfied:

1. ***Size Reduction Condition:*** *For all $1 \leq i < j \leq n$:*

$$\mu_{j,i} \in \mathcal{A},$$

and

$$N(\mu_{j,i}) \leq \mathcal{M}(\Lambda) < 1,$$

2. ***Lovász Condition:*** *For all $2 \leq i \leq n$, the following inequality holds:*

$$\|\mathbf{b}_i(i)\|^2_{\mathcal{A}/K} \geq (\delta - N(\mu_{i,i-1}))\|\mathbf{b}_{i-1}(i-1)\|^2_{\mathcal{A}/K}$$

where $\delta \in (\mathcal{M}(\Lambda), 1]$ is the reduction quality parameter.

Algorithm 1 QLLL Reduction

$\mathbf{B}_{red}, \mathbf{U} \leftarrow QLLL(\mathbf{B}, \Lambda, \delta)$:

$\mathbf{R} \leftarrow QR(\mathbf{B})$ ▷ Alg. 3
$\mathbf{U} \leftarrow \mathbf{I}_{n \times n}$
$k \leftarrow 2$
while $k \leq n$ **do**
  $\mathbf{R}, \mathbf{U} \leftarrow SizeRed(\mathbf{R}, \mathbf{U}, \Lambda, k)$ ▷ Alg. 2
  if $\delta \cdot r^2_{k-1,k-1} > \|r_{k-1,k}\|^2_{\mathcal{A}/K} + r^2_{k,k}$ **then**
    $SwapVectors(\mathbf{R}, \mathbf{U}, k)$ ▷ Alg. 8
    $k \leftarrow \max(2, k-1)$
  else ▷ Lovász condition fulfilled
    $k \leftarrow k + 1$
  end if
end while
$\mathbf{B}_{red} = \mathbf{BU}$
return $\mathbf{B}_{red}, \mathbf{U}$

Algorithm 2 Size Reduction

$SizeReduce(\mathbf{R}, \mathbf{U}, \Lambda, k)$:

for $i = (k-1), ..., 1$ **do**
  $\mu_{k,i} \leftarrow r_{i,k} \cdot r_{i,i}^{-1}$
  if $Q_\Lambda(\mu_{k,i}) \neq 0$ **then**
    $\mathbf{r}_k \leftarrow \mathbf{r}_k - Q_\Lambda(\mu_{k,i}) \cdot \mathbf{r}_i$
    $\mathbf{u}_k \leftarrow \mathbf{u}_k - Q_\Lambda(\mu_{k,i}) \cdot \mathbf{u}_i$
  end if
end for

3.2 QR Decomposition

To implement LLL efficiently we store the Gram-Schmidt coefficients in the upper-triangular $\mathbf{R}$ matrix of a QR decomposition, given by Algorithm 3. In fact, we are able to solely use a transform matrix and this matrix $\mathbf{R} \in \mathcal{A}^{n \times n}$ during our LLL algorithm. The Gram-Schmidt Orthogonalisation of $\mathbf{B} \in \mathcal{A}^{m \times n}$ is the tuple $(\mathbf{d}, \mathbf{M})$, where $d_i = \|\mathbf{b}_i(i)\|_{\mathcal{A}/K} \in K$ and $m_{i,j} = \mu_{j,i} \in \mathcal{A}$ on and above the diagonal and zero otherwise.

Algorithm 3 Quaternion Gram-Schmidt Decomposition

$\mathbf{R} \leftarrow QR(\mathbf{B})$:

$\mathbf{Q} \leftarrow \mathbf{B}$
for $i \leftarrow 1$ to n **do**
  for $j \leftarrow 1$ to $i-1$ **do**
    $\mathbf{q}_i \leftarrow \mathbf{q}_i - \langle \mathbf{q}_j, \mathbf{b}_i \rangle_{\mathcal{A}} \cdot \|\mathbf{q}_j\|_{\mathcal{A}/K}^{-2} \cdot \mathbf{q}_j$
  end for
end for
$r_{i,j} \leftarrow \langle \mathbf{q}_i, \mathbf{b}_j \rangle_{\mathcal{A}} \cdot \|\mathbf{q}_i\|_{\mathcal{A}/K}^{-1}$ if $i \leq j$, 0 otherwise
return R

Let $\mathbf{B} \in \mathcal{A}^{m \times n}$ be the current basis matrix, then there exists a unique Gram-Schmidt decomposition $\mathbf{B} = \mathbf{QR}$ where $\mathbf{Q} \in \mathcal{A}^{m \times n}$ is orthogonal, and $\mathbf{R} \in \mathcal{A}^{n \times n}$ uppertriangular with positive definite diagonal called the R-factor. Such a decomposition can be computed in $\mathcal{O}(dn^2)$ operations. This R-factor has the following form:

$$\mathbf{R} = \mathrm{diag}(\mathbf{d})\mathbf{M}, \qquad r_{i,j} = d_i \mu_{ij} = \frac{\langle \mathbf{b}_j, \mathbf{b}_i(i) \rangle_{\mathcal{A}}}{\|\mathbf{b}_i(i)\|_{\mathcal{A}/K}} \in \mathcal{A},$$

so in particular $r_{i,i} = \|\mathbf{b}_i(i)\|_{\mathcal{A}/K}$, and $\mu_{j,i}$ may be recovered by $r_{i,j} \cdot r_{i,i}^{-1}$.

4 LLL over Non-Euclidean Quaternion Orders

4.1 Moving to an Overlattice

As described, we are only able to run quaternion LLL on lattices over Euclidean orders. We wish to extend this to natural orders of rational quaternion algebras that are contained in Euclidean maximal orders. To overcome this non-Euclidean nature, we propose lifting our lattice into an overlattice, and then reducing over a different Euclidean (and maximal) order, before returning back to our original lattice. Of course, when considering rational quaternion algebras, we are restricted to the algebras $\mathcal{A}_\iota$ as described in Subsection 2.4 and their respective orders, $\mathcal{N}_\iota \subset \Lambda_\iota \subset \mathcal{A}_\iota$.

This gives us the following setup: Let $\mathcal{L}$ be a $\mathcal{N}_\iota$-lattice with basis $\mathbf{B}$, and $\mathcal{L}'$ a Λ_ι-lattice also given by $\mathbf{B}$; then $\mathcal{L} \subset \mathcal{L}'$. We are then able to run QLLL over Λ_ι on $\mathbf{B}$ to obtain a QLLL-reduced basis $\mathbf{B}'$ for $\mathcal{L}'$, though note that in general, this is *not* a basis for $\mathcal{L}$.

This now presents the problem to find a unimodular transform $\mathbf{U}$ such that $\mathbf{B}'\mathbf{U}$ recovers a reduced basis for $\mathcal{L}$.

4.2 LLL and a Sublattice Approach

We first begin by describing the naïve approach, Algorithm 4, which also recovers the shortest basis vector of our solutions given, though this does not return a basis of the lattice that was given. This approach uses the fact that the natural order is of course contained within the maximal order by our construction, and one can take any element lying in the maximal order and multiply by two to recover an element in the natural order.

Whilst we are both guaranteed that we are left with vectors in the natural order that are linearly independent and to some degree reduced, we no longer have a set of vectors that is spanned over the natural order; our set of vectors is no longer a basis.

Proposition 2. *Suppose that $\mathcal{L}$ is a lattice spanned over $\mathcal{N}_\iota$ with basis $\mathbf{b}_1, ..., \mathbf{b}_n \in \mathcal{A}^n$, and $\mathcal{L}'$ is the respective overlattice of $\mathcal{L}$ when the basis is spanned over Λ_ι. Let $\mathbf{b}'_1, ..., \mathbf{b}'_n$ denote the basis $\mathbf{b}_1, ..., \mathbf{b}_n$ when QLLL reduced over Λ_ι.*

Suppose that k is the smallest index such that $\mathbf{b}'_k \notin \mathcal{L}$. Then let $\mathcal{L}^\dagger$ be the lattice given by the basis $\mathbf{v}_1, ..., \mathbf{v}_n := \mathbf{b}'_1, ..., \mathbf{b}'_{k-1}, 2\mathbf{b}'_k, ..., 2\mathbf{b}'_n$ spanned over $\mathcal{N}_\iota$, where $\mathcal{L}^\dagger \subset \mathcal{L}$, with $[\mathcal{L} : \mathcal{L}^\dagger] = 1$ if all $\mathbf{b}'_i \in \mathcal{L}$ or $[\mathcal{L} : \mathcal{L}^\dagger] = 2^{n-k+1}$ otherwise and we have

$$\frac{\|\mathbf{v}_i\|^2}{\lambda_i(\mathcal{L})^2} \leq \frac{1}{(\delta - \mathcal{M}(\Lambda_\iota))^n},\ 1 \leq i \leq k-1,$$
$$\frac{\|\mathbf{v}_i\|^2}{\lambda_i(\mathcal{L})^2} \leq \frac{4}{(\delta - \mathcal{M}(\Lambda_\iota))^n},\ k \leq i \leq n.$$

Proof. The result follows from the fact that for each $\mathbf{b}'_i \in \mathcal{L}' \setminus \mathcal{L}$, $2\mathbf{b}'_i \in \mathcal{L}$ (for any $x \in \mathbb{Z}$, $2x \in \mathbb{Z}$ also, and for any half integer, multiplying by two moves it into $\mathbb{Z}$), and the standard bounds relating to LLL reduced bases vector lengths. □

Algorithm 4 QLLL and a Sublattice

$\mathbf{B}_{red} \leftarrow$ sublattice-$QLLL(\mathbf{B}_{\mathcal{I}}, \mathcal{A}_k, \delta)$:

1: $\mathbf{B}' \leftarrow QLLL(\mathbf{B}_{\mathcal{I}}, \Lambda_k, \delta)$
2: $l \leftarrow \text{argmin}(\mathbf{b}'_i \notin \mathcal{N})$
3: $\mathbf{B}_{red} \leftarrow \{\mathbf{b}'_1, ..., \mathbf{b}'_{l-1}, 2\mathbf{b}'_l, ..., 2\mathbf{b}'_{n-1}, 2\mathbf{b}'_n\}$
4: **return** $\mathbf{B}_{red}$

4.3 LLL and Recovering a True Basis

In practice, we do not use LLL to truly solve any kind of cryptographic SVP problem, and its polynomial runtime instead lends itself to use as a preprocessing step; often BKZ assumes an LLL reduced basis as input. Because of this, the prior approach may not be attractive as it cannot be used in the manner that LLL is typically used.

To resolve this dilemma, we propose an alternative approach that relies on the observation that given two vectors, we are able to guarantee that both are moved back into the natural order using only unimodular operations which we then extend to the n-dimensional case.

Lemma 2. *Let $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{A}_\iota^n$ be vectors such that*

$$\mathbf{v}_1 = \sum_{i=1}^{n} x_i \mathbf{b}_i, \qquad \mathbf{v}_2 = \sum_{i=1}^{n} y_i \mathbf{b}_i,$$

with $x_i, y_i \in \Lambda_\iota$ and $\mathbf{b}_i \in \mathcal{A}_\iota^n$. Then there exists a $\mathbf{U} \in GL_2(\mathcal{N}_\iota)$ such that $\mathbf{U}(\mathbf{v}_1, \mathbf{v}_2)^T = (\mathbf{w}_1, \mathbf{w}_2)^T$, where

$$\mathbf{w}_1 = \sum_{i=1}^{n} \alpha_i \mathbf{b}_i, \qquad \mathbf{w}_2 = \sum_{i=1}^{n} \beta_i \mathbf{b}_i,$$

with $\alpha_i, \beta_i \in \mathcal{N}_\iota$ and $\|\mathbf{w}_i\| \leq F_{i+n}^{1/2} \max_j \|\mathbf{v}_j\|$, where F_i is the i-th Fibonacci number.

Proof. We assume first that $n = 2$. Then

$$\mathbf{v}_1 = x_1\mathbf{b}_1 + x_2\mathbf{b}_2, \qquad \mathbf{v}_2 = y_1\mathbf{b}_1 + y_2\mathbf{b}_2.$$

We may assume without loss of generality that $x_1, y_1 \in \mathcal{N}_\iota$, as otherwise we may multiply $\mathbf{v}_i$ by some unit such that this holds.

If $(x_1, x_2) \in \mathcal{N}_\iota^2$, then the top row of $\mathbf{U}$ is set to be $(1, 0)$. Otherwise, suppose $x_2 \notin \mathcal{N}_\iota$. If $y_2 \notin \mathcal{N}_\iota$, then $x_2 \pm y_2 \in \mathcal{N}_\iota$. Note that at least one of $\|\mathbf{v}_1 \pm \mathbf{v}_2\|^2 \leq \|\mathbf{v}_1\|^2 + \|\mathbf{v}_2\|^2$, so we set $\mathbf{w}_1$ to be the lesser of these two vectors, so that $\|\mathbf{w}_1\|^2 \leq \|\mathbf{v}_1\|^2 + \|\mathbf{v}_2\|^2 \leq 2 \max_j \|\mathbf{v}_j\|^2$.

Suppose now that $y_2 \in \mathcal{N}$. We let $x_2 = s + t\omega$, for some $s \in \mathcal{N}$, $t \in \mathbb{Z}$ that is not even, and $\omega \in \Lambda_\iota \backslash \mathcal{N}_\iota$.[2] If $s = 0$, then $x_2\omega \notin \mathcal{N}_\iota$, and so $\omega(x_1, x_2) \pm \omega(y_1, y_2) \in \mathcal{N}_\iota^2$, so we set $\mathbf{w}_1$ to be the lesser of $\omega(\mathbf{v}_1 \pm \mathbf{v}_2)$ in length. If $s \neq 0$, then $\bar{\omega} x_2 \notin \mathcal{N}_\iota$, and so we set $\mathbf{w}_1$ to be the lesser of $\bar{\omega}\mathbf{v}_1 \pm \omega\mathbf{v}_2$ in length.

In the same way, after setting $\mathbf{v}_1 \to \mathbf{w}_1$ and fixing $\mathbf{v}_2$ (which is a unimodular transform), we may then choose a unimodular transform to set $\mathbf{v}_2 \to \mathbf{w}_2$ with the pair $(\mathbf{w}_1, \mathbf{v}_2)$ so that $\|\mathbf{w}_2\|^2 \leq \|\mathbf{w}_1\|^2 + \|\mathbf{v}_2\|^2 \leq 3 \max_j \|\mathbf{v}_j\|^2$.

In the case where n is some arbitrary integer, this process may be repeated iteratively to force all the $x_i \to \alpha_i, y_i \to \beta_i \in \mathcal{N}_\iota$, where if we denote $\mathbf{w}_1^{(k)}, \mathbf{w}_2^{(k)}$ to be the vectors obtained when repeating this process k times, in the next step we get $\|\mathbf{w}_1^{(k+1)}\|^2 \leq \|\mathbf{w}_1^{(k)}\|^2 + \|\mathbf{w}_2^{(k)}\|^2$. Hence, the appearance of the Fibonacci number in the bound becomes apparent.

□

Theorem 1 (Naturally Quaternion LLL Reduced). *Let $\mathcal{L}$ be an $\mathcal{N}_\iota$-lattice of even*[3] *rank given by basis $\mathbf{B}$. Given $\mathbf{B}'$, the output of QLLL over Λ_ι with input $\mathbf{B}$, there exists a unimodular transform that sends $\mathbf{B}'$ to a basis of $\mathcal{L}$, $\mathbf{V}$ such that*

$$\|\mathbf{v}_i\|^2 \leq F_{n+2-i'} \|\mathbf{b}'_{2\lceil i/2 \rceil}\|^2$$

where $i' \equiv i \mod 2$, and we call such a basis naturally-QLLL reduced.

Proof. We begin by taking the QLLL reduced basis $\mathbf{B}'$, for which we consider disjoint pairwise to get tuples of basis vectors, $(\mathbf{b}'_{2i-1}, \mathbf{b}'_{2i})$ for $i \in \{1, ..., n/2\}$. These tuples may be expressed with respect to the basis $\mathbf{B}$, and we may now apply Lemma 2 to each of these pairs, and we call the output of this $\mathbf{V}$. Since $\mathbf{B}$ is a $\mathcal{N}_\iota$-lattice, the output of $\mathbf{V}$ is therefore also a basis of a $\mathcal{N}_\iota$-lattice since the coefficients of the basis vectors are all in $\mathcal{N}_\iota$. We may also directly apply the output vector bound from Lemma 2, where clearly the bound is always with respect to the vector of the tuple with largest index; the Fibonacci term is dependent on which of the tuple we are addressing. We are able to do this without concern of the rest of the basis exactly because these pairs are disjoint, and so we may consider them independently whilst we effectively fix all others.

□

[2] Intuitively, ω is the "half-integer" part.

[3] We note that this can be extended to full generality though we will restrict to the even case since it aids analysis of bounds, as well as being true for cryptographically relevant cases such as those arising from power-of-two cyclotomics.

4.4 Modified BKZ with Blocksize 2

Our last 'LLL' proposal aims to address the issue of maintaining a basis in $\mathcal{N}_\iota$ within the algorithm itself, running as a subroutine alongside the size reduction step. Because of this, though, we are unable to use LLL as the Lovász condition places too heavy a restriction on the pairwise vectors, as well as needing stronger notions of reduction. As such, we look to BKZ with blocksize two using Gauss reduction alongside the observation used in the above section.

Definition 16. *Let $\mathbf{b}_1, \mathbf{b}_2 \in \mathcal{A}^n$. Then we say $\mathbf{b}_1, \mathbf{b_2}$ are Gauss reduced with respect to the order $\mathcal{O} \subset \mathcal{A}$ if, for all $q \in \mathcal{O}$,*

$$\|\mathbf{b}_1\| \leq \|\mathbf{b}_2\| \leq \|\mathbf{b}_2 + q\mathbf{b}_1\|.$$

Algorithm 5 Gauss Reduction

$\mathbf{b}_1', \mathbf{b}_2' \leftarrow \mathit{GaussReduce}(\mathbf{b}_1, \mathbf{b}_2, \mathcal{O})$
1: $B_1 \leftarrow \|\mathbf{b}_1\|^2$
2: $\mu \leftarrow \frac{\langle \mathbf{b}_1, \mathbf{b}_2 \rangle}{B_1}$
3: $\mathbf{b_2} \leftarrow \mathbf{b}_2 - Q_{\mathcal{O}}(\mu)\mathbf{b}_1$
4: $B_2 \leftarrow \|\mathbf{b}_2\|^2$
5: **while** $B_2 < B_1$ **do**
6: Swap $\mathbf{b}_1, \mathbf{b}_2$
7: $B_1 \leftarrow B_2$
8: $\mu \leftarrow \frac{\langle \mathbf{b}_1, \mathbf{b}_2 \rangle}{B_1}$
9: $\mathbf{b_2} \leftarrow \mathbf{b}_2 - Q_{\mathcal{O}}(\mu)\mathbf{b}_1$
10: $B_2 \leftarrow \|\mathbf{b}_2\|^2$
11: **end while**
12: **return** $\mathbf{b}_1, \mathbf{b}_2$

We note here that we are able to define Gauss reduction with respect to an arbitrary order $\mathcal{O}$, where intuitively, this can be thought of as minimising an element by subtracting elements of the order. We will need to assume that the order is maximal to make further statements about the quality of the output however.

Theorem 2. *Let $\mathbf{b}_1, \mathbf{b}_2$ be an output of Gauss reduction with respect to the Euclidean order Λ_1, then $\|\mathbf{b}_1\| = \lambda_1, \|\mathbf{b}_2\| = \lambda_2$.*

Proof. Since we have a Gauss reduced basis, we have that

$$\begin{aligned} &\|\mathbf{b}_2 + q\mathbf{b}_1\|^2 \geq \|\mathbf{b}_2\|^2 \geq \|\mathbf{b}_1\|^2 \\ \Leftrightarrow\ &N(\mu - q) \geq N(\mu) \\ &\Rightarrow N(\mu) \leq \mathcal{M}(\Lambda_\iota), \end{aligned} \tag{1}$$

and we have by definition and (1),

$$\begin{aligned}\|\mathbf{b}_2(2)\|^2 &= \|\mathbf{b}_2\|^2 - N(\mu)\|\mathbf{b}_1\|^2 \\ &\geq (1 - N(\mu))\|\mathbf{b}_2\|^2 \\ &\geq (1 - \mathcal{M}(\Lambda_\iota))\|\mathbf{b}_2\|^2. \end{aligned} \tag{2}$$

Now, let us consider an arbitrary vector in our lattice as a combination of our basis vectors for $x, y \in \Lambda_\iota$,

$$\begin{aligned}\|x\mathbf{b}_1 + y\mathbf{b}_2\|^2 &= N(y)\|\mathbf{b}_2(2)\|^2 + N(x + \mu y)\|\mathbf{b}_1(1)\|^2 \\ &\geq N(y)(1 - \mathcal{M}(\Lambda_\iota))\|\mathbf{b}_2\|^2\end{aligned}$$

by using (2).

If $N(y) = 1$, then y is a unit and so without loss of generality, we have the left hand side becoming $\|x\mathbf{b}_1 + \mathbf{b}_2\|^2$, which by our original assumption is greater than $\|\mathbf{b}_2\|^2$ since it is Gauss reduced.

Now, suppose that y is not a unit, and so $N(y) \geq 2$. When $\Lambda_\iota = \Lambda_1$, we have

$$\begin{aligned}\|x\mathbf{b}_1 + y\mathbf{b}_2\|^2 &\geq N(y)(1 - 1/2)\|\mathbf{b}_2\|^2 \\ &\geq \|\mathbf{b}_2\|^2.\end{aligned}$$

□

Theorem 3. *Given a rank two quaternion $\mathcal{N}_1$-lattice, $\mathcal{L}$ with basis $\mathbf{b}_1, \mathbf{b}_2$, and the Gauss reduced basis $\mathbf{v}_1, \mathbf{v}_2$ of the Euclidean Λ_1-overlattice $\mathcal{L}'$, then one can find a vector $\mathbf{w} \in \mathcal{L}$ such that $\|\mathbf{w}\| = \lambda_1(\mathcal{L})$ in constant time.*

Proof. From Theorem 2, we know that $\|\mathbf{v}_1\| = \lambda_1(\mathcal{L}'), \|\mathbf{v}_2\| = \lambda_2(\mathcal{L}')$. In the case that $\mathbf{v}_1 \in \mathcal{L}$, we are done since $\mathcal{L} \subset \mathcal{L}'$, and $\mathbf{w} := \mathbf{v}_1$.

Now, let us consider when $\mathbf{v}_1 \notin \mathcal{L}$, but $\mathbf{v}_2 \in \mathcal{L}$. Suppose that there exists a vector $\mathbf{w} \in \mathcal{L}$, where $\mathbf{w} \neq \mathbf{v}_1$ such that $\|\mathbf{w}\| < \|\mathbf{v}_2\|$. Then since $\mathcal{L} \subset \mathcal{L}'$, we have that $\mathbf{w} \in \mathcal{L}'$ and so $\lambda_1(\mathcal{L}') < \|\mathbf{w}\| < \lambda_2(\mathcal{L}')$ and either $\mathbf{w} = \mathbf{v}_1$ or $\|\mathbf{v}_2\| \neq \lambda_2(\mathcal{L}')$ both contradictions. As such, we are simply able to return $\mathbf{v}_2$ as the shortest lattice vector of $\mathcal{L}$.

The last case to consider is when neither $\mathbf{v}_i$ are in $\mathcal{L}$. Rewriting the gauss reduced vectors in terms of the original basis vectors, we get

$$\mathbf{v}_1 = x_1\mathbf{b}_1 + x_2\mathbf{b}_2, \qquad \mathbf{v}_2 = y_1\mathbf{b}_1 + y_2\mathbf{b}_2$$

for $x_i, y_i \in \Lambda_1$.

As in earlier sections, we recall the fact that for any vector in $\mathcal{L}'$, we have that two times this vector is in $\mathcal{L}$. From this, we have

$$\lambda_1(\mathcal{L}') \leq \lambda_1(\mathcal{L}) \leq 2\lambda_1(\mathcal{L}').$$

This then means that for a vector in $\mathcal{L}$ to be the shortest lattice vector, it must satisfy

$$\|x\mathbf{v}_1 + y\mathbf{v}_2\|^2 \leq 4\|\mathbf{v}_1\|^2$$

for $x, y \in \Lambda_1$, and again considering the Gram Schmidt decomposition,

$$N(y)\|\mathbf{v}_2(2)\|^2 \geq \frac{N(y)}{2}\|\mathbf{v}_2\|^2$$
$$N(y) \leq 8.$$

This leaves us with a constant number of elements y that we must enumerate in order to find $\mathbf{w}$.

□

Definition 17 (Quaternion BKZ-Reduction). *Let $\mathcal{A}$ be a totally definite quaternion algebra over a number field K, and $\mathbf{B}$ the basis for the rank n lattice $\mathcal{L}$ over an order $\mathcal{O} = \mathcal{N}$ or $\mathcal{O} = \Lambda$. We say that $\mathbf{B}$ is β-BKZ reduced if for all $i \in \{1, ..., n-1\}$,*

$$\|\mathbf{b}_i(i)\| = \lambda_1(\pi_i(\mathcal{L}_{[i:min(i+\beta-1,n)]}))$$

and $N(\mu_{i,j}) < \mathcal{M}(\mathcal{O})$ for all $1 \leq j < i \leq n$; are size reduced with respect to the order.

Using Gauss reduction over the Euclidean order before lifting this to a solution for the shortest vector in the original projected sublattice with Theorem 3, we can replace a standard SVP oracle call giving a modified 2-BKZ algorithm, Algorithm 9 (Appendix C).

Theorem 4. *Given an $\mathcal{N}_1$-lattice $\mathcal{L}$ with basis $\mathbf{B}$, Algorithm 9 returns a basis $\mathbf{B}'$ that is 2-QBKZ reduced.*

Proof. We have restricted ourselves to the $\mathcal{A} = (\frac{-1,-1}{\mathbb{Q}})$ setting, and so we have that $\mathcal{M}(\mathcal{N}_1) = 1$. This means that whilst the size reduction step is not guaranteed to decrease the norm of the Gram-Schmidt vector, it cannot *increase* it. Our insertion step is a strictly decreasing function with respect to the Gram-Schmidt vector norms, and so may only happen a finite number of times; coupled with the fact that size reduction is non-increasing, we are therefore guaranteed termination.

We begin by showing that $\mathbf{B}'$ satisfies the SVP condition with respect to the rank-2 projected sublattice. Since the algorithm has terminated, we must have for all $i \in \{1, ..., n\}$ that $\|\mathbf{b}_i(i)\| \leq \|\mathbf{v}\| = \lambda_1(\pi_i(\mathcal{L}_{[i,i+1]}))$. Since $\mathbf{b}_i(i) \in \pi_i(\mathcal{L}_{[i,i+1]})$, we therefore must have $\mathbf{b}_i(i) = \lambda_1(\pi_i(\mathcal{L}_{[i,i+1]}))$ as required. It can clearly be seen that since we size reduce after every change, the output basis is also therefore size reduced, and since as stated before this does not increase the Gram-Schmidt vector lengths, it does not impact the SVP condition.

□

5 Extending to Arbitrary Blocksizes

Definition 18 (Quaternion HKZ-Reduction). *Let $\mathcal{A}$ be a totally definite quaternion algebra over a number field K, and $\mathbf{B}$ the basis for the rank n lattice*

$\mathcal{L}$ over an order $\mathcal{O} = \mathcal{N}$ or $\mathcal{O} = \Lambda$. We say that $\mathbf{B}$ *is HKZ reduced if for all* $i \in \{1, ..., n-1\}$,

$$\|\mathbf{b}_i(i)\| = \lambda_1(\pi_i(\mathcal{L}_{[i:n]}))$$

and $N(\mu_{i,j}) < \mathcal{M}(\mathcal{O})$ *for all* $1 \leq j < i \leq n$*; are size reduced with respect to the order.*

Lemma 3. *Let* $\mathbf{B}$ *be a QHKZ reduced basis of an* $\mathcal{O}$*-lattice,* $\mathcal{L}$*. Then*

$$\frac{1}{\mathcal{M}(\mathcal{O})(i-1)+1}\lambda_i(\mathcal{L})^2 \leq \|\mathbf{b}_i\|^2 \leq (\mathcal{M}(\mathcal{O})(i-1)+1)\lambda_i(\mathcal{L})^2.$$

Proof. We begin by remarking that $\lambda_1(\pi_i(\mathcal{L}_{[i:n]})) \leq \lambda_i(\mathcal{L})$, and so we have that $\|\mathbf{b}_i(i)\| \leq \lambda_i(\mathcal{L})$. To obtain an upperbound, we first consider the Gram-Schmidt decomposition;

$$\begin{aligned}
\|\mathbf{b}_i\|^2 &= \|\mathbf{b}_i(i)\|^2 + \sum_{j=1}^{i-1} N(\mu_{i,j})\|\mathbf{b}_j(j)\|^2 \\
&\leq \|\mathbf{b}_i(i)\|^2 + \mathcal{M}(\mathcal{O})\sum_{j=1}^{i-1}\|\mathbf{b}_j(j)\|^2 \\
&\leq \lambda_i(\mathcal{L})^2 + \mathcal{M}(\mathcal{O})\sum_{j=1}^{i-1}\lambda_j(\mathcal{L})^2 \\
&\leq \lambda_i(\mathcal{L})^2 + \mathcal{M}(\mathcal{O})(i-1)\cdot\lambda_i(\mathcal{L})^2 \\
&\leq (\mathcal{M}(\mathcal{O})(i-1)+1)\cdot\lambda_i(\mathcal{L})^2.
\end{aligned}$$

We now turn to a lower bound, and observe that for $j \leq i$, $\|\mathbf{b}_j(j)\|^2 \leq \|\mathbf{b}_i\|^2$. Again, considering the Gram-Schmidt decomposition, we have

$$\begin{aligned}
\|\mathbf{b}_j\|^2 &\leq \|\mathbf{b}_j(j)\|^2 + \mathcal{M}(\mathcal{O})\sum_{k=1}^{j-1}\|\mathbf{b}_k(k)\|^2 \\
&\leq (\mathcal{M}(\mathcal{O})(i-1)+1)\|\mathbf{b}_i\|^2 \\
\Rightarrow \lambda_i(\mathcal{L})^2 &\leq (\mathcal{M}(\mathcal{O})(i-1)+1)\|\mathbf{b}_i\|^2 \\
\Rightarrow \|\mathbf{b}_i\|^2 &\geq \frac{1}{(\mathcal{M}(\mathcal{O})(i-1)+1)}\lambda_i(\mathcal{L})^2.
\end{aligned}$$

□

Theorem 5. *Let* $\mathcal{L}$ *(*$\mathcal{L}'$ *resp.) be a rank-n* $\mathcal{N}_\iota$*-lattice (*Λ_ι*-lattice resp.) where* $\mathcal{L}, \mathcal{L}'$ *are defined by the same basis,* $\mathbf{B}$*. Then, given* $\mathbf{B}'$*, an HKZ reduced basis for* $\mathcal{L}'$*, one can recover a vector* $\mathbf{v} \in \mathcal{L}$ *such that* $\|\mathbf{v}\| = \lambda_1(\mathcal{L})$ *in* $2^{O(n)}$ *time.*

Proof. As before, we have the following relationship of successive minima of $\mathcal{L}, \mathcal{L}'$:

$$\lambda_1(\mathcal{L}') \leq \lambda_1(\mathcal{L}) \leq [\Lambda_\iota : \mathcal{N}_\iota] \cdot \lambda_1(\mathcal{L}').$$

Set $A = [\Lambda_\iota : \mathcal{N}_\iota]^2 \|\mathbf{b}'_1\|^2$, so that any $\mathbf{v} \in \mathcal{L}$ achieving $\lambda_1(\mathcal{L})$ satisfies $\|\mathbf{v}\|^2 \leq A$.

Expressing $\mathbf{v} = \sum_{i=1}^{n} x_i \mathbf{b}'_i$ with $x_i \in \Lambda_\iota$, the Gram-Schmidt decomposition gives

$$\|\mathbf{v}\|^2 = \sum_{i=1}^{n} N\left(x_i + \sum_{j>i} \mu_{j,i}\, x_j\right) \|\mathbf{b}'_i(i)\|^2 \leq A.$$

Since $\mathbf{B}'$ is HKZ reduced, size-reduction gives $N(\mu_{i+1,i}) \leq \mathcal{M}(\Lambda_\iota)$, and so

$$\|\mathbf{b}'_{i+1}(i+1)\|^2 \geq (1 - \mathcal{M}(\Lambda_\iota))\|\mathbf{b}'_i(i)\|^2.$$

By induction, $\|\mathbf{b}'_i(i)\|^2 \geq (1 - \mathcal{M}(\Lambda_\iota))^{i-1}\|\mathbf{b}'_1\|^2$ for all i.

Consider the highest nonzero coordinate, i.e. the largest i with $x_i \neq 0$ (so that $x_{i+1} = ... = x_n = 0$). For this coordinate, the GSO coefficient is exactly $c_i = x_i$, contributing $N(x_i)\|\mathbf{b}'_i(i)\|^2$ to $\|\mathbf{v}\|^2$.

If $N(x_i) \geq 2$, then

$$2\|\mathbf{b}'_i(i)\|^2 \leq A = [\Lambda_\iota : \mathcal{N}_\iota]^2 \|\mathbf{b}'_1\|^2,$$

so $\|\mathbf{b}'_i(i)\|^2 \leq \frac{[\Lambda_\iota : \mathcal{N}_\iota]^2}{2}\|\mathbf{b}'_1\|^2$. By the decay bound, $\|\mathbf{b}'_i(i)\|^2 \geq (1-\mathcal{M}(\Lambda_\iota))^{i-1}\|\mathbf{b}'_1\|^2$, giving

$$(1 - \mathcal{M}(\Lambda_\iota))^{i-1} \leq \frac{[\Lambda_\iota : \mathcal{N}_\iota]^2}{2}.$$

This is satisfied only for $i \leq i_0$, where

$$i_0 = 1 + \left\lfloor \frac{\log([\Lambda_\iota : \mathcal{N}_\iota]^2/2)}{\log(1/(1 - \mathcal{M}(\Lambda_\iota)))} \right\rfloor.$$

For Λ_1 $i_0 = 1 + \lfloor \log_2 2 \rfloor = 2$, for Λ_2 $i_0 = 1 + \lfloor \log_3 8 \rfloor = 2$.

Therefore, for all $i > i_0$: the highest nonzero coordinate x_i must satisfy $N(x_i) = 1$; x_i is a unit of Λ_ι. The unit group $\Lambda_\iota^\times$ is finite; for Λ_1, $|\Lambda_1^\times| = 24$ and for Λ_2, $|\Lambda_2^\times| = 12$.

We enumerate $(x_n, x_{n-1}, ..., x_1)$ top-down, maintaining the remaining budget $B_i = A - \sum_{j>i} l_j$ where $l_j = N(c_j)\|\mathbf{b}'_j(j)\|^2$. At each level i:

- If $i > i_0$ and $x_{i+1} = ... = x_n = 0$: the unit constraint gives $x_i \in \{0\} \cup \Lambda_\iota^\times$, at most $|\Lambda_\iota^\times| + 1$ candidates.
- If $i > i_0$ and some $x_j \neq 0$ for $j > i$: we enumerate x_i satisfying the norm bound $N(c_i) \leq B_i/\|\mathbf{b}'_i(i)\|^2$, which by the Lemma above also restricts x_i to at most $O(|\Lambda_\iota^\times|)$ candidates (since $N(x_i + \text{centre})$ is bounded by a constant depending on $\mathcal{M}$ and the budget).
- If $i \leq i_0$: the norm bound gives $N(c_i) \leq B_i/\|\mathbf{b}'_i(i)\|^2 \leq A/\|\mathbf{b}'_i(i)\|^2 = O(1)$, giving $O(1)$ candidates.

A branch is pruned whenever $B_i < 0$.

The enumeration tree has at most $(|\Lambda_\iota^\times| + 1)^n$ nodes, giving $2^{O(n)}$ total cost (each node is processed in $O(n)$ arithmetic operations).

Among all completed coefficient vectors $(x_1, ..., x_n)$, we check $\mathcal{L}$-membership by verifying that the coordinates in the original basis $\mathbf{B}$ lie in $\mathcal{N}_\iota$ (cost $O(n^2)$ per vector). The shortest such vector is returned. Since $A \geq \lambda_1(\mathcal{L})^2$, the search ball contains the shortest vector of $\mathcal{L}$, and the unit constraint does not exclude it (any $\mathbf{v}$ with $\|\mathbf{v}\|^2 \leq A$ and highest nonzero coordinate at $i > i_0$ necessarily has $N(x_i) = 1$ by the argument above).

□

Algorithm 6 Modified Quaternion β-BKZ

$\mathbf{B}_{red} \leftarrow \beta\text{-}QBKZ(\mathbf{B}, \delta))$:

$\Omega = 0$
while $\Omega < n$ **do**
  for $i = 1, ..., n$ **do**
    $\beta' \leftarrow \min(i + \beta - 1, n)$
    $\mathbf{v}_{\Lambda_1} = x_i\pi_i(\mathbf{b}_i) + ... + x_{i+\beta'}\pi_i(\mathbf{b}_{i+\beta'}) \leftarrow HKZ(\pi_i(\mathbf{B}_{[i,i+\beta']}), \Lambda_1)$
    $\mathbf{v} \in \mathcal{N}_1 \leftarrow$ Theorem 5 using $(\mathbf{v}_{\Lambda_1})$
    if $\|\mathbf{v}\| < \|\mathbf{b}_i(i)\|$ **then**
      $\mathbf{v}' \leftarrow x_i\mathbf{b}_i + ... + x_{i+\beta'}\mathbf{b}_{i+\beta'}$
      Insert $\mathbf{v}'$ at index i
      Remove linear dependencies
      for $1 \leq j \leq i - 1$ **do**
        if $N(\mu_{i,i-j}) > \mathcal{M}(\mathcal{N}_1)$ **then**
          $\mathbf{b}_i \leftarrow \mathbf{b}_i - Q_{\mathcal{N}_1}(\mu_{i,i-j})\mathbf{b}_{i-j}$
        end if
      end for
      $\Omega \leftarrow 0$
    else
      $\Omega \leftarrow \Omega + 1$
    end if
  end for
end while

Remark 3. We note that using the same multiplication by two method as Subsection 4.2, we recover an approximate SVP oracle with constant approximation factor.

Theorem 6. *Given an $\mathcal{O}$-lattice $\mathcal{L}$ with basis $\mathbf{B}$, Algorithm 6 returns a basis $\mathbf{B}'$ that is β-QBKZ reduced.*

Proof. The proof follows exactly as in Theorem 4.

6 Output Bounds and Complexity Analysis

We begin by recounting the general bounds for shortest basis vector, orthogonality defect, and time complexity for Algorithm 1, QLLL, reducing a full-rank n-dimensional lattice basis over the Euclidean order Λ_ι as given in [35] (adapted for the consideration of the order and for $\mathcal{A}$ over $\mathbb{Q}$) and we remark that $\operatorname{disc}(\mathcal{N}/\mathbb{Z}) = [\Lambda : \mathcal{N}]^2 \cdot \operatorname{disc}(\Lambda/\mathbb{Z})$.

First basis vector:	$\|\mathbf{b_1}\|^2 \leq \left(\frac{1}{\delta - \mathcal{M}(\Lambda_\iota)}\right)^{\frac{n-1}{2}} \frac{\operatorname{vol}(\mathcal{L})^{2/n}}{\lvert\operatorname{disc}(\Lambda_\iota/\mathbb{Z})\rvert^{1/n}}$
Orthogonality Defect:	$\Omega_{\mathcal{L}}(\mathbf{B}) \leq \frac{1}{\lvert\operatorname{disc}(\Lambda_\iota/\mathbb{Z})\rvert^{1/2}} \left(\frac{1}{\delta - \mathcal{M}(\Lambda_\iota)}\right)^{\frac{n(n-1)}{4}}$
Asymptotic Complexity:	$\mathcal{O}(n^4 \log_{\frac{1}{\delta}}(n))$

The relevant discriminants and indices are collected in Appendix A.

6.1 Sublattice Approach

We begin again with the simplest approach to analyse, the sublattice approach of Subsection 4.2. Let $\mathcal{L}$ be a rank-n $\mathcal{N}_\iota$-lattice with basis $\mathbf{B}$, and $\mathcal{L}'$ the overlattice obtained by spanning $\mathbf{B}$ over $\Lambda_\iota \supset \mathcal{N}_\iota$. QLLL over Λ_ι produces a reduced basis $\mathbf{V}$ of $\mathcal{L}'$. If k is the smallest index with $\mathbf{v}_k \notin \mathcal{L}$, the sublattice $\mathcal{L}^\dagger \subset \mathcal{L}$ has basis $\mathbf{v}_1, ..., \mathbf{v}_{k-1}, 2\mathbf{v}_k, ..., 2\mathbf{v}_n$, i.e., $\mathcal{L}^\dagger \subset \mathcal{L} \subset \mathcal{L}'$.

Since in the worst case, the first basis vector is not in $\mathcal{N}_\iota^n$, we must therefore multiply it by two, giving

$$\begin{aligned}\|\mathbf{b}_1\|^2 &\leq \frac{4}{\lvert\operatorname{disc}(\Lambda_\iota/\mathbb{Z})\rvert^{1/n}} \left(\frac{1}{\delta - \mathcal{M}(\Lambda_\iota)}\right)^{\frac{n-1}{2}} \operatorname{vol}(\mathcal{L}')^{2/n} \\ &= \frac{4}{\lvert\operatorname{disc}(\Lambda_\iota/\mathbb{Z})\rvert^{1/n}} \left(\frac{1}{\delta - \mathcal{M}(\Lambda_\iota)}\right)^{\frac{n-1}{2}} \left(\frac{\operatorname{vol}(\mathcal{L})}{[\Lambda_\iota : \mathcal{N}_\iota]}\right)^{2/n}.\end{aligned}$$

Considering the orthogonality defect, we have

$$\begin{aligned}\Omega_{\mathcal{L}^\dagger}(\mathbf{B}) &= \frac{\prod_{j=1}^n \|\mathbf{b}_j\|}{\operatorname{vol}(\mathcal{L}^\dagger)} \\ &= \frac{2^{n-k+1} \prod_{j=1}^n \|\mathbf{v}_j\|}{2^{n-k+1} \cdot \operatorname{vol}(\mathcal{L})} \\ &= \frac{\prod_{j=1}^n \|\mathbf{v}_j\|}{[\Lambda_\iota : \mathcal{N}_\iota] \operatorname{vol}(\mathcal{L}')} \\ &= \Omega_{\mathcal{L}'}(\mathbf{V}) / [\Lambda_\iota : \mathcal{N}_\iota] \\ &\leq \frac{1}{\sqrt{\lvert\operatorname{disc}(\mathcal{N}/\mathbb{Z})\rvert}} \left(\frac{1}{\delta - \mathcal{M}(\Lambda_\iota)}\right)^{\frac{n(n-1)}{4}}.\end{aligned}$$

Finally, our post-processing is a linear in n number of constant time operations ($k \leq n$ multiplications by 2), leaving the asymptotic complexity unchanged.

6.2 Naturally QLLL Reduced

Again, let $\mathbf{B}$ be the final outputted basis of the $\mathcal{N}_\iota$-lattice $\mathcal{L}$, and $\mathbf{V}$ be the intermediary QLLL reduced basis of the Λ_ι-lattice $\mathcal{L}'$.

Directly from the definition, we have the bounds

$$\|\mathbf{b}_1\|^2 \leq F_{n+1}\|\mathbf{v}_2\|^2, \qquad \text{and} \qquad \|\mathbf{v}_2\|^2 \leq \left(\frac{1}{\delta - \mathcal{M}(\Lambda_\iota)}\right)^{n-1} \lambda_2(\mathcal{L}')^2.$$

Suppose that $\mathbf{w} = x_1\mathbf{v}_1 + ... + x_n\mathbf{v}_n$ is a shortest vector in $\mathcal{L}$, for some $x_i \in \Lambda_\iota$. Here, there are two options — either $x_2 = ... = x_n = 0$, in which case $\mathbf{w}$ is already the shortest vector in $\mathcal{L}$, in which case $\|\mathbf{v}_1\| = \lambda_1(\mathcal{L})$, so $\mathbf{b}_1 = \omega\mathbf{v}_1 \in \mathcal{L}$ is a shortest vector for some appropriately chosen unit $\omega \in \Lambda$, or $x_i \neq 0$ for some $2 \leq i$, in which case $\|\mathbf{v}_i(i)\| \leq \lambda_1(\mathcal{L})$, and so

$$\|\mathbf{b}_1\|^2 \leq F_{n+1}\|\mathbf{v}_2\|^2 \leq F_{n+1}\left(\frac{1}{\delta - \mathcal{M}(\Lambda_\iota)}\right)^{n-1} \lambda_1(\mathcal{L})^2,$$

and so we can always bound $\|\mathbf{b}_1\|/\lambda_1(\mathcal{L})$.

This leads us to an upper bound on the ratio $\|\mathbf{b}_1\|^2/\mathrm{vol}(\mathcal{L})^{2/n}$ - we denote by

$$\gamma_{n,\mathcal{O}} := \max_{\mathcal{L}_\mathcal{O}} \frac{\lambda_1(\mathcal{L}_\mathcal{O})^2}{\mathrm{vol}(\mathcal{L}_\mathcal{O})^{2/n}},$$

where the maximum is taken over all rank n lattices spanned over $\mathcal{O}$, and $\lambda_1(\mathcal{L}_\mathcal{O})$ is the first successive minimum of this lattice. This value is known to be upper bounded in value. Using this, we have

$$\|\mathbf{b}_1\|^2 \leq F_{n+1}\left(\frac{1}{\delta - \mathcal{M}(\Lambda_\iota)}\right)^{n-1} \gamma_{n,\mathcal{N}_\iota}\mathrm{vol}(\mathcal{L})^{2/n}.$$

We remark that $\gamma_{n,\mathcal{O}} \leq \frac{\gamma_{4n}}{2} \cdot |\mathrm{disc}(\mathcal{O}/\mathbb{Z})|^{1/4-1/n}$ for any order $\mathcal{O}$, where γ_k is the rational Hermite constant rank k.

$$\Rightarrow \|\mathbf{b}_1\|^2 \leq \gamma_{4n} \cdot F_{n+1}/2\left(\frac{1}{\delta - \mathcal{M}(\Lambda_\iota)}\right)^{n-1} \frac{|\,\mathrm{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{1/4}}{|\,\mathrm{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{1/n}}\mathrm{vol}(\mathcal{L})^{2/n}.$$

To understand the upper bounds on the orthogonality defect, we begin by looking at the denominator:

$$\begin{aligned}
\|\mathbf{b}_i\| &\leq \sqrt{F_{n+2-i'}}\|\mathbf{v}_{2\lceil i/2\rceil}\| \\
\Rightarrow \prod_{i=1}^{n}\|\mathbf{b}_i\| &\leq (\sqrt{F_{n+1}\cdot F_{n+2}})^{n/2}\prod_{i=1}^{n/2}\|\mathbf{v}_{2i}\|^2 \\
\Rightarrow \Omega_{\mathcal{L}}(\mathbf{B}) &\leq \frac{(F_{n+1}F_{n+2})^{n/4}}{[\Lambda_\iota : \mathcal{N}_\iota]}\cdot\frac{\prod_{i=1}^{n/2}\|\mathbf{v}_{2i}\|^2}{\mathrm{vol}(\mathcal{L}')} \\
&\leq \frac{(F_{n+1}F_{n+2})^{n/4}}{[\Lambda_\iota : \mathcal{N}_\iota]}\cdot\frac{\prod_{i=1}^{n/2}\|\mathbf{v}_{2i}\|}{\prod_{i=1}^{n/2}\|\mathbf{v}_{2i-1}\|}\cdot\Omega_{\mathcal{L}'}(\mathbf{V}) \\
&\leq \frac{(F_{n+1}F_{n+2})^{n/4}}{[\Lambda_\iota : \mathcal{N}_\iota]}\cdot\left(\frac{1}{\delta-\mathcal{M}(\Lambda_\iota)}\right)^{n/2}\cdot\Omega_{\mathcal{L}'}(\mathbf{V}) \\
&\leq \frac{(F_{n+1}F_{n+2})^{n/4}}{|\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{1/2}}\cdot\left(\frac{1}{\delta-\mathcal{M}(\Lambda_\iota)}\right)^{\frac{n(n+1)}{4}}.
\end{aligned}$$

We note that whilst the upper bound on the first basis vector length is worse with this subroutine, we would like to draw attention to the following facts:

1. The upper bound assumes that every element of the vector is not in the natural order, and so must be moved. In practice, this will not be the case - in fact the first vector could even begin in the natural order. The Fibonacci coefficient assumes this worst case; as a factor that is exponential in n, assuming this worst case makes the bound significantly worse.
2. This is further compounded by the fact that we must take an upper bound on (a variant of) Hermite's Constant. We note that in practice, the output almost certainly outperforms this generous upper bound.
3. Additionally, in practice LLL is used as a preprocessing step/subroutine. When this is the case, it is often important that certain structures are preserved. In our case here, we maintain all quaternion structure, and also note that the orthogonality defect upper bound is still much better; this is arguably a more important metric for BKZ input where a well-conditioned profile is strongly valued.

In terms of complexity, we have in the worst case that when considering a pair of vectors

$$\mathbf{v}_1 = x_1\mathbf{b}_1 + ... + x_n\mathbf{b}_n, \qquad \mathbf{v}_2 = y_1\mathbf{b}_1 + ... + y_n\mathbf{b}_n,$$

all $x_i, y_i \notin \mathcal{N}_\iota$. If this is so, we must do our naturalisation process $n+1$ times, and so across our whole basis when applying our full process disjoint pairwise, we have $(n(n+1))/2$ additions/multiplications. When expressing the complexity asymptotically with big O notation, this does not affect the final expression.

6.3 Comparison to LLL and Complex LLL

Suppose that an adversary is given a rank-n $\mathcal{N}_\iota$-lattice, $\mathcal{L}$. We now compare the output quality based on the bounds we have derived of the following approaches: the baseline embedding of $\mathcal{L}$ as a $\mathbb{Z}$-lattice before using standard LLL/BKZ, embedding as a complex lattice before using complex LLL, the sublattice reduction as in Subsection 4.2, natural QLLL reduction as in Subsection 4.3, and quaternion BKZ as in Subsection 5. We also remark that by our definition of the volume, we are able to take a "volume normalised" comparison. We will also take $\delta = 1$ for ease of analysis and as is often the taken arbitrariliy close to one in practice, and give the comparison of these bounds in Table 1, and plot these in Figures 1, 2 in Appendix B. We remark that our subroutines can be viewed as trading vector length for better orthogonality (in the worst case), whilst maintaining the better asymptotic complexity given by QLLL.

6.4 Quaternion BKZ

We follow the standard proof of bounds for a BKZ reduced basis. Recall that by definition, we have

$$\lambda_1(\mathcal{L}) \leq \gamma_{n,\mathcal{O}}^{1/2} \cdot \mathrm{vol}(\mathcal{L})^{1/n},$$

and so for a QBKZ reduced basis,

$$\begin{aligned}
\|\mathbf{b}_i(i)\| &= \lambda_1(\pi_i(\mathcal{L}_{[i:\min(n,i+\beta-1)]})) \\
&\leq \gamma_{\beta,\mathcal{N}_\iota}^{1/2} \cdot |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{1/2\beta} \left(\prod_{j=i}^{i+\beta-1} \|\mathbf{b}_j(j)\| \right)^{1/\beta} \\
\Rightarrow \|\mathbf{b}_i(i)\|^\beta &\leq \gamma_{\beta,\mathcal{N}_\iota}^{\beta/2} \cdot |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{1/2} \left(\prod_{j=1}^{i+\beta-1} \|\mathbf{b}_j(j)\| \right),
\end{aligned}$$

for all $i \in [1, ..., n-\beta+1]$. Multiplying all such inequalities, we obtain

$$\begin{aligned}
\prod_{i=1}^{n-\beta+1} \|\mathbf{b}_i(i)\|^\beta &\leq \prod_{i=1}^{n-\beta+1} \gamma_{\beta,\mathcal{N}_\iota}^{\beta/2} \cdot |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{1/2} \left(\prod_{j=i}^{i+\beta-1} \|\mathbf{b}_j(j)\| \right) \\
\Rightarrow \prod_{i=1}^{\beta-1} \|\mathbf{b}_i(i)\|^{\beta-i} &\leq \gamma_{\beta,\mathcal{N}_\iota}^{\frac{\beta(n-\beta+1)}{2}} \cdot |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{\frac{n-\beta+1}{2}} \\
&\quad \cdot \prod_{i=1}^{\beta-1} \|\mathbf{b}_{n-\beta+i+1}(n-\beta+i+1)\|^{\beta-i}. \qquad (3)
\end{aligned}$$

We also have that for any $k < \beta$, the projected sublattice of the block $\mathbf{B}_{[1:k]}$ is also SVP reduced, and so we also have

$$\|\mathbf{b}_1(1)\|^k \leq \gamma_{\beta,\mathcal{N}_\iota}^{k/2} \cdot |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{1/2} \left(\prod_{j=1}^{k} \|\mathbf{b}_j(j)\| \right),$$

for $k \in [2, ..., \beta - 1]$. Again, multiplying all these inequalities, we have

$$\begin{aligned}
\|\mathbf{b}_1(1)\|^{\frac{\beta(\beta-1)}{2}} &\leq |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{(\beta-2)/2} \left(\prod_{i=2}^{\beta-1} \gamma_{i,\mathcal{N}_\iota}^{i/2}\right) \prod_{i=1}^{\beta-1} \|\mathbf{b}_i(i)\|^{\beta-i} \\
&\leq |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{(\beta-2)/2} \left(\prod_{i=2}^{\beta-1} \gamma_{\beta,\mathcal{N}_\iota}^{\beta/2}\right) \prod_{i=1}^{\beta-1} \|\mathbf{b}_i(i)\|^{\beta-i} \\
&\leq |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{(\beta-2)/2} \left(\gamma_{\beta,\mathcal{N}_\iota}^{\frac{\beta(\beta-2)}{2}}\right) \prod_{i=1}^{\beta-1} \|\mathbf{b}_i(i)\|^{\beta-i}. \qquad (4)
\end{aligned}$$

Note that the products of Gram-Schmidt vectors on the left hand side of eq. (3) and right hand side of eq. (4) are the same. Multiplying (3) and (4) then gives

$$\|\mathbf{b}_1(1)\|^{\frac{\beta(\beta-1)}{2}} \leq |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{\frac{n-1}{2}} \cdot \gamma_{\beta,\mathcal{N}_\iota}^{\frac{\beta(n-1)}{2}} \cdot \prod_{i=1}^{\beta-1} \|\mathbf{b}_{n-\beta+i+1}(n-\beta+i+1)\|^{\beta-i}$$

We now use the fact that $\|\mathbf{b}_i(i)\| = \lambda_1(\pi_i(\mathcal{L}_{[i:\min(n,i+\beta-1)]})) \leq \lambda_1(\mathcal{L})$ to bound the product and obtain

$$\begin{aligned}
\|\mathbf{b}_1(1)\|^{\frac{\beta(\beta-1)}{2}} &\leq |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{\frac{n-1}{2}} \cdot \gamma_{\beta,\mathcal{N}_\iota}^{\frac{\beta(n-1)}{2}} \lambda_1(\mathcal{L})^{\frac{\beta(\beta-1)}{2}} \\
\Rightarrow \|\mathbf{b}_1(1)\| = \|\mathbf{b}_1\| &\leq |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{\frac{n-1}{\beta(\beta-1)}} \cdot \gamma_{\beta,\mathcal{N}_\iota}^{\frac{n-1}{\beta-1}} \lambda_1(\mathcal{L}) \\
&\leq |\operatorname{disc}(\mathcal{N}_\iota/\mathbb{Z})|^{\frac{n-1}{\beta(\beta-1)}} \gamma_{\beta,\mathcal{N}_\iota}^{\frac{n-1}{\beta-1}+1/2} \operatorname{vol}(\mathcal{L})^{\frac{1}{n}}.
\end{aligned}$$

A comparison to BKZ can be seen in Appendix B, Table 2 and Figure 3; we remark that in practice, such algorithms outperform these bounds significantly.

7 Applying QuaILLL to Rank-2 Module-LIP over Cyclotomic Fields

7.1 Constructing Quaternion-Ideal Lattice Bases

Let ζ_m denote a primitive m-th root of unity, and $F = \mathbb{Q}(\zeta_m)$ be its cyclotomic field. Let $K = \mathbb{Q}(\zeta_m + \zeta_m^{-1})$ be the maximal totally real subfield of F, so that $F = K(\sqrt{a})$, following the setup of [6].

Now, let $\mathcal{A} = (\frac{a,-1}{K})$ be a totally definite quaternion algebra since K is totally real and $a \in K$ is totally negative. We denote by $\mathcal{O}_K$ and $\mathcal{O}_\mathcal{A}$ maximal orders of K and $\mathcal{A}$ respectively and let $\mathcal{N}_\mathcal{A}$ be the natural order, $\mathcal{N}_\mathcal{A} := \mathcal{O}_K + i\mathcal{O}_K + j\mathcal{O}_K + k\mathcal{O}_K$. Further, let $\mathbb{H}_a = (\frac{a,-1}{\mathbb{R}})$, and let $\mathcal{N}_{\mathbb{H}_a}$ be the natural order of $(\frac{a,-1}{\mathbb{Q}})$.

Now, let us consider a principal left ideal $\mathcal{I} \subseteq \mathcal{O}_\mathcal{A}$, that is given by the use of Algorithm 10. We wish to embed this ideal into a so-called quaternion lattice, and give an embedding in such a way to reduce dimension compared to

the standard use of the canonical embedding whilst maintaining quaternionic structure. This embedding assumes $\mathcal{O}_\mathcal{A} = \mathcal{N}_\mathcal{A}$; the ramifications of which are addressed in Subsection 7.2.

Let $\sigma_i : K \to \mathbb{R}$ denote the usual canonical embedding (we note that this is to $\mathbb{R}$ as K is totally real), and further, we define the quaternionic canonical embedding where for $q \in \mathcal{A}$, we have

$$\sigma_i^{\mathbb{H}_a} : \mathcal{A} \to \mathbb{H}_a, \qquad \sigma_i^{\mathbb{H}}(q) := \sigma_i(q_1) + \sigma_i(q_2)i + \sigma_i(q_3)j + \sigma_i(q_4)k,$$

$$\Sigma^{\mathbb{H}_a} : \mathcal{A} \to \mathbb{H}_a^{n\times n}, \qquad \Sigma^{\mathbb{H}}(q) := \begin{pmatrix} \sigma_1^{\mathbb{H}}(q) & \sigma_n^{\mathbb{H}}(q) & \dots & \sigma_2^{\mathbb{H}}(q) \\ \sigma_2^{\mathbb{H}}(q) & \sigma_1^{\mathbb{H}}(q) & \dots & \sigma_3^{\mathbb{H}}(q) \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_n^{\mathbb{H}}(q) & \sigma_{n-1}^{\mathbb{H}}(q) & \dots & \sigma_1^{\mathbb{H}}(q) \end{pmatrix}.$$

Since $\mathcal{I}$ is a principal ideal, it is generated by some $\alpha \in \mathcal{A}$ such that $\mathcal{I} = \alpha\mathcal{O}_\mathcal{A}$, but to express this as an $\mathcal{N}_{\mathbb{H}_a}$-lattice, we need a mapping

$$\alpha\mathcal{O}_\mathcal{A} \mapsto \Sigma^{\mathbb{H}}(\alpha)\mathbf{B}\mathcal{N}_{\mathbb{H}_a}^n$$

where $\mathbf{B}$ is the matrix shaping $\mathcal{I}$ as an $\mathcal{N}_{\mathbb{H}_a}$-module, analogously to the standard real embedding of an ideal of a number field into a $\mathbb{Z}$-lattice.

However, we do not have $\mathcal{I}$ in this form, instead, as the intersection of two fractional ideals

$$\mathcal{I} = \mathfrak{b}_1^{-1}\mathcal{O}_\mathcal{A} \cap q\mathfrak{b}_2^{-1}\mathcal{O}_\mathcal{A},$$

where $\mathfrak{b}_i$ are fractional ideals of $\mathcal{O}_F$, and $q \in \mathcal{A}$ [6]. As such, we will continue considering these as $\mathfrak{c}_1, \mathfrak{c}_2$, fractional ideals of $\mathcal{O}_A$ respectively. We are then able to consider this as an ideal with two generators, c_i, c_i', and such generators are efficient to compute [38].[4]

Since we have assumed the order $\mathcal{O}_\mathcal{A}$ is in fact the natural order, $\mathcal{O}_\mathcal{A} = \mathcal{O}_K \oplus i\mathcal{O}_K \oplus j\mathcal{O}_K \oplus k\mathcal{O}_K$, where $\mathcal{O}_K = \omega_1\mathbb{Z} + \dots + \omega_n\mathbb{Z}$, with $\omega_i \in K$.

Briefly expressing $\mathcal{O}_\mathcal{A}$ in terms of this new $\mathbb{Z}$-basis, we get

$$\begin{aligned} \mathcal{O}_\mathcal{A} &= (\omega_1\mathbb{Z} + \dots + \omega_n\mathbb{Z}) \oplus i(\omega_1\mathbb{Z} + \dots + \omega_n\mathbb{Z}) \\ &\qquad \oplus j(\omega_1\mathbb{Z} + \dots + \omega_n\mathbb{Z}) \oplus k(\omega_1\mathbb{Z} + \dots + \omega_n\mathbb{Z}) \\ &= \omega_1(\mathbb{Z} \oplus i\mathbb{Z} \oplus j\mathbb{Z} \oplus k\mathbb{Z}) \oplus \dots \oplus \omega_n(\mathbb{Z} \oplus i\mathbb{Z} \oplus j\mathbb{Z} \oplus k\mathbb{Z}) \\ &= \omega_1\mathcal{N}_{\mathbb{H}_a} \oplus \dots \oplus \omega_n\mathcal{N}_{\mathbb{H}_a}. \end{aligned}$$

We argue that this assumption on the maximality of the natural order of $\mathcal{O}_\mathcal{A}$ is one that is valid since we are interested in two cases; when m is a power of two or a power of three. When in the power of three case, this maximality assumption holds, and for powers of two the natural order is very close to being maximal. This means that we have moved into a sublattice, but have that this sublattice is very close to the lattice that we originally wanted.

[4] We remark that since we find a generator matrix where the dimension must be reduced anyway, an arbitrary number of generators may be used in this step.

From here, we can apply our embedding $\Sigma^{\mathbb{H}_a}$ to each of the generators of $\mathfrak{c}_i$, and concatenating these matrices, denoting this by $\Sigma^{\mathbb{H}_a}(c_i, c_i')$, giving us a generator matrix for the the lattice $\mathcal{L}_{\mathfrak{c}_i}$.

To recover a basis matrix from a generator we must now put the generator in its Hermite Normal Form (HNF), which then once we remove the n all zero columns at the end will give us a basis matrix. This can be done in polynomial time, though we note that it must be done over a Euclidean order. This is an issue that we will be faced with and address in the following sections, so for now we will simply note that we will compute the HNF of the generator matrix over the corresponding Euclidean order after embedding, before moving back to our lattice over the natural order as a subroutine.

$$\mathbf{B}_{\mathfrak{c}_i} = \mathrm{HNF}(\Sigma^{\mathbb{H}_a}(c_i, c_i'))\sigma(\omega) \in \mathbb{H}_a^{n\times n}$$

where $\omega = (\omega_1, ..., \omega_n)$ and n is the degree of K over $\mathbb{Q}$.

Now, having two bases of the fractional-ideal-lattices, we can simply compute their intersection using their duals:

$$\mathbf{B}_{\mathfrak{c}_1\cap\mathfrak{c}_2} = (\mathrm{HNF}(\mathbf{B}_{\mathfrak{c}_1}^{\wedge}|\mathbf{B}_{\mathfrak{c}_2}^{\wedge}))^{\wedge}$$

where $\mathbf{B}_{\mathfrak{c}_i}^{\wedge} = \mathbf{B}_{\mathfrak{c}_i}(\mathbf{B}_{\mathfrak{c}_i}^H\mathbf{B}_{\mathfrak{c}_i})^{-1}$ is the basis for the dual lattice, again taking the HNF to recover a basis.

This then gives us a basis to represent the $\mathcal{N}_{\mathbb{H}_a}$-lattice of $\mathcal{I}$, and is described in Algorithm 11 (Appendix C);

$$\mathbf{B}_{\mathcal{I}} = \mathbf{B}_{\mathfrak{c}_1\cap\mathfrak{c}_2} \in \mathbb{H}^{n\times n}.$$

We follow the approach of [6] to construct a quaternion nrdPIP problem, and then Algorithm 11 to construct a basis for a lattice. Our goal is now to solve SVP in this lattice, and we now describe how our algorithm can be used against certain structured LIP instances. We will now further justify our initial claim that we assume $\mathcal{O}_{\mathcal{A}}$ is the natural order.

Lemma 4 (Corollary B.17, [6]). *Let $F = \mathbb{Q}(\zeta_m)$ and K its totally real subfield. Then for $\mathcal{A} = (\frac{a,b}{K})$, the natural order of $\mathcal{A}$, $\mathcal{N}_{\mathcal{A}}$ is related to the maximal order, $\mathcal{O}_{\mathcal{A}}$ in the following way:*

1. *If $m = 2^c$ (with $c > 2$), then $\mathcal{N}_{\mathcal{A}} \subsetneq \mathcal{O}_{\mathcal{A}}$;*
2. *If $m = p^c$ or $2p^c$, then*
   (a) *if $p \equiv 1 \mod 4$ then $\mathcal{N}_{\mathcal{A}} \subsetneq \mathcal{O}_{\mathcal{A}}$;*
   (b) *if $p \equiv 3 \mod 4$ then $\mathcal{N}_{\mathcal{A}}$ is maximal;*
3. *Otherwise, $\mathcal{N}_{\mathcal{A}}$ is maximal.*

We now consider cases 1 and 2(a) from Lemma 4, when the natural order $\mathcal{N}_{\mathcal{A}} = \mathcal{O}_F \oplus \mathcal{O}_F j \subseteq \mathcal{O}_{\mathcal{A}}$. As such, given some principal ideal of $\mathcal{O}_{\mathcal{A}}$, $\mathcal{I} = \alpha\mathcal{O}_{\mathcal{A}}$ then we can define $\mathcal{I}' = \alpha\mathcal{N}_{\mathcal{A}} \subseteq \alpha\mathcal{O}_{\mathcal{A}}$. Since the natural order is very close to maximal, this new ideal $\mathcal{I}'$ is dense in $\mathcal{I}$ as a sublattice. Whilst the solution to nrdPIP for $\mathcal{I}$ may not exist in $\mathcal{I}'$, we are able to solve an approximate γ-nrdPIP

in $\mathcal{I}'$ using one of the three algorithms given prior, and where γ depends on $[\mathcal{O}_\mathcal{A} : \mathcal{N}_\mathcal{A}]$.

With this in mind, we end up with the following cases that we are able to address with our algorithms, and for each, we denote how close the natural order is to maximal:

- $m = 2^c$ (with $c > 2$); $[\mathcal{O}_\mathcal{A} : \mathcal{N}_\mathcal{A}] = 4$
- $4|m$, but $m \neq p^c$ or $2p^c$; $\mathcal{N}_\mathcal{A}$ is maximal
- $3|m$; $\mathcal{N}_\mathcal{A}$ is maximal

In Algorithms 10, 11, 7, assume that m falls into one of these cases, and we remark that for the final two cases, the algorithms and bounds given earlier may directly apply.

For completeness, we now describe how to apply this to power-of-two cyclotomics; as-is it may apply to conductors $4|m$, $3|m$ directly, exactly the cases that are not covered by [37, 24].

Algorithm 7 QuaILLL Applied to Rank-2 MLIP

$F = \mathbb{Q}(\zeta_m)$,
$G = (\mathbf{G} = \mathbf{B}^*\mathbf{B}, \mathfrak{a}_1, \mathfrak{a}_2)$
$G' = (\mathbf{G}, \mathfrak{b}_1, \mathfrak{b}_2) \cong G$
δ
) :

$\mathcal{I}, \mathcal{A} \leftarrow Q_Ideal(F, G, G')$
if $3|m$ **then**
    $\Lambda \leftarrow \Lambda_2$
else if $4|m$ **then**
    $\Lambda \leftarrow \Lambda_1$
else
    return $\perp$
end if
$\mathbf{B}_\mathcal{I} \leftarrow Q_Ideal_Lattice(\mathcal{I})$
$\mathbf{B}'_\mathcal{I} \leftarrow QuaILLL(\mathbf{B}_\mathcal{I}, \Lambda, \delta)$ ▷ Use any one of the algorithms described in Sections 4.2, 4.3, 5

7.2 Power-of-Two Cyclotomics and Hawk

We now address the most challenging case; that of power-of-two cyclotomics, as in the case of Hawk. As established in the previous subsection, power-of-two conductors fall into the category in which $\mathcal{N}_\mathcal{A}$ is not maximal, and so we end up working in a sublattice. In the same flavour as [23], we would like to then be able to make claims about the approximation factor we have when we lift our solution in the sublattice back to the desired setting.

We begin by recapping our setting: Let $F = \mathbb{Q}(\zeta_{2^c})$, and $\mathcal{A} = \left(\frac{-1,-1}{K}\right)$, where K is the maximal totally real subfield of F. Further, let $\mathcal{O}_\mathcal{A}$ be a maximal order of $\mathcal{A}$ such that it contains the natural order $\mathcal{N}_\mathcal{A}$. Let $\mathcal{I} = a\mathcal{O}_\mathcal{A}$, and $\mathcal{J} = a\mathcal{N}_\mathcal{A}$. Then, we have that

$$[\mathcal{O}_\mathcal{A} : \mathcal{N}_\mathcal{A}] = [\mathcal{I} : \mathcal{J}] = \frac{\mathrm{vol}(\mathcal{L}_\mathcal{J})}{\mathrm{vol}(\mathcal{L}_\mathcal{I})},$$

where $\mathcal{L}_\mathcal{I}, \mathcal{L}_\mathcal{J}$ are the lattices obtained by our embedding $\mathcal{I}, \mathcal{J}$ respectively. We have this since the canonical embedding preserves covolume ratios. In our setting as described above, we get that $[\mathcal{O}_\mathcal{A} : \mathcal{N}_\mathcal{A}] = 4$.

On top of this, since $\mathcal{J} \subset \mathcal{I}$, $\mathcal{L}_\mathcal{J}$ is a sublattice of $\mathcal{L}_\mathcal{I}$, thus it follows that any solution in our sublattice is a solution of the lattice we really care about.

Suppose that we have obtained a reduced basis $\mathbf{B}$ of $\mathcal{L}_\mathcal{J}$ as in the output of Algorithm 7. Since $\mathcal{L}_\mathcal{J} \subset \mathcal{L}_\mathcal{I}$, every vector of $\mathcal{L}_\mathcal{J}$ is automatically a vector of $\mathcal{L}_\mathcal{I}$, so the output basis is valid for the larger lattice. The index $[\mathcal{I} : \mathcal{J}] = [\mathcal{O}_\mathcal{A} : \mathcal{N}_\mathcal{A}] = 4$ introduces the following additional factors when interpreting $\mathcal{L}_\mathcal{J}$-bounds as $\mathcal{L}_\mathcal{I}$-bounds.

Any bound of the form $\|\mathbf{b}_1\|^2 \leq \gamma^2 \cdot \mathrm{vol}(\mathcal{L}_\mathcal{J})^{2/n}$ translates to

$$\|\mathbf{b}_1\|^2 \leq \gamma^2 \cdot \left([\mathcal{I} : \mathcal{J}] \cdot \mathrm{vol}(\mathcal{L}_\mathcal{I})\right)^{2/n} = \gamma^2 \cdot 4^{2/n} \cdot \mathrm{vol}(\mathcal{L}_\mathcal{I})^{2/n},$$

since $\mathrm{vol}(\mathcal{L}_\mathcal{J}) = [\mathcal{I} : \mathcal{J}] \cdot \mathrm{vol}(\mathcal{L}_\mathcal{I})$. The HSVP approximation factor for $\mathcal{L}_\mathcal{I}$ is therefore $\gamma' = 4^{1/n} \cdot \gamma$; for large n, this penalty is mild.

Since $\mathcal{L}_\mathcal{J} \subset \mathcal{L}_\mathcal{I}$, we have $\lambda_i(\mathcal{L}_\mathcal{I}) \leq \lambda_i(\mathcal{L}_\mathcal{J}) \leq [\mathcal{I} : \mathcal{J}] \cdot \lambda_i(\mathcal{L}_\mathcal{I})$. A bound $\|\mathbf{b}_i\|^2 / \lambda_i(\mathcal{L}_\mathcal{J})^2 \leq C$ translates to

$$\frac{\|\mathbf{b}_i\|^2}{\lambda_i(\mathcal{L}_\mathcal{I})^2} = \frac{\|\mathbf{b}_i\|^2}{\lambda_i(\mathcal{L}_\mathcal{J})^2} \cdot \frac{\lambda_i(\mathcal{L}_\mathcal{J})^2}{\lambda_i(\mathcal{L}_\mathcal{I})^2} \leq C \cdot [\mathcal{I} : \mathcal{J}]^2 = 16C.$$

The orthogonality defect of $\mathbf{B}$ as a basis of $\mathcal{L}_\mathcal{I}$ is

$$\Omega_{\mathcal{L}_\mathcal{I}}(\mathbf{B}) = \frac{\prod \|\mathbf{b}_i\|}{\mathrm{vol}(\mathcal{L}_\mathcal{I})} = [\mathcal{I} : \mathcal{J}] \cdot \frac{\prod \|\mathbf{b}_i\|}{\mathrm{vol}(\mathcal{L}_\mathcal{J})} = [\mathcal{I} : \mathcal{J}] \cdot \Omega_{\mathcal{L}_\mathcal{J}}(\mathbf{B}) = 4\,\Omega_{\mathcal{L}_\mathcal{J}}(\mathbf{B}).$$

In all three metrics, the index introduces a constant factor depending only on $[\mathcal{O}_\mathcal{A} : \mathcal{N}_\mathcal{A}]$:

Metric	Additional factor
First basis vector (HSVP)	$[\mathcal{O}_\mathcal{A} : \mathcal{N}_\mathcal{A}]^{1/n} = 4^{1/n}$
Successive minima	$[\mathcal{O}_\mathcal{A} : \mathcal{N}_\mathcal{A}]^2 = 16$
Orthogonality defect	$[\mathcal{O}_\mathcal{A} : \mathcal{N}_\mathcal{A}] = 4$

These are independent of the choice of subroutine (sublattice, naturally QLLL, or quaternion BKZ) and of n. In particular, the improvements of the quaternion approaches over $\mathbb{Z}$-LLL (as quantified in Table 1) are preserved up to these constant factors.

8 Conclusion

We believe that this is the first instance of lattice reduction over quaternion rings applied in a cryptographic setting, and we further generalise the notions introduced by [26] to provide solutions for non-Euclidean quaternion orders. We have also generalised a number of lattice reduction definitions to the quaternion setting, though we would also like to note that these can be further generalised to arbitrary Euclidean integer rings (notably Gauss reduction, HKZ, and BKZ).

These culminate in the description of a quaternion BKZ algorithm. Further heuristic and numerical analysis of which we leave for future work, though would be interested in its comparison to BKZ over the corresponding $\mathbb{Z}$-lattice and module BKZ, through the extension of the Geometric Series Assumption, as well as profile simulation. This will allow analysis of comparative blocksize requirements and rate of convergence. Such numerical analysis would also lend itself to the analysis of using approximate SVP oracles, as briefly mentioned in Remark 3.

These results are then motivated by and applied to the ideal lattice arising from the quaternion PIP problems related to rank-2 module LIP, specifically those from cyclotomics such as Hawk, though more broadly applicable to cyclotomics with conductors m where $3|m$ or $4|m$. To do so, we provide the description of an embedding that maintains quaternion structure for us to use during reduction — how the ideal structure can also be used is a more general question that is also ripe for the picking in future literature. One hurdle not quite overcome by this embedding is the appearance of the maximal order in these quaternion algebras; classification of maximal orders of relevant algebras in terms of an $\mathcal{N}$-basis would greatly help and is another direction for future work. The work [24] may also allow our algorithms to be utilised in a quaternionic setting; we must note that the trace-zero part is not an order, and so our work may not be directly applied, as well as not solving exact-SVP.

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A Quaternion Orders

A.1 Quantisation Functions

For $a = a_1 + a_2 i + a_3 j + a_4 k \in (\frac{-1,-1}{\mathbb{R}})$, the quantisation into Λ_1 proceeds as follows:

$$Q_{\Lambda_1}(a) \;=\; \min_{N(\cdot)} \begin{cases} \lfloor a_1 \rceil \;+\; \lfloor a_2 \rceil i \;+\; \lfloor a_3 \rceil j \;+\; \lfloor a_4 \rceil k \\ \sum_{\ell=1}^{4} \left(\lfloor a_\ell - \frac{1}{2} \rceil + \frac{1}{2} \right) e_\ell \end{cases},$$

where $e_1 = 1, e_2 = i, e_3 = j, e_4 = k$, and picking the minimal case with respect to the norm, where $\lfloor \cdot \rceil$ denotes rounding to the nearest integer. Thus, Q_{Λ_1} maps each quaternion to the nearest Hurwitz quaternion,[5] either in the integer lattice $\mathbb{Z}^4$ or in the shifted lattice $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}) + \mathbb{Z}^4$.

We also give the quantisation function for Λ_2, where for $a = a_1 + a_2 i + a_3 j + a_4 k \in (\frac{-1,-3}{\mathbb{R}})$,

$$Q_{\Lambda_2}(a) \;=\; \min_{N(\cdot)} \begin{cases} Q_{\Lambda_1}(a) \\ \lfloor a_1 \rceil \;+\; \left(\lfloor a_2 - \frac{1}{2} \rceil + \frac{1}{2} \right) i \;+\; \left(\lfloor a_3 - \frac{1}{2} \rceil + \frac{1}{2} \right) j \;+\; \lfloor a_4 \rceil k \\ \left(\lfloor a_1 - \frac{1}{2} \rceil + \frac{1}{2} \right) \;+\; \lfloor a_2 \rceil i \;+\; \lfloor a_3 \rceil j \;+\; \left(\lfloor a_4 - \frac{1}{2} \rceil + \frac{1}{2} \right) k \end{cases}$$

again, taking whichever is smallest with respect to the norm.

[5] We note that Λ_1 are the Hurwitz quaternions, and $(\frac{-1,-1}{\mathbb{R}}) = \mathbb{H}$ the Hamiltonian quaternions.

A.2 Order Constants

We collect the Euclidean minima [4], discriminants, and indices for the orders used in this work. The natural order of a quaternion algebra $\mathcal{A} = \left(\frac{a,b}{\mathbb{Q}}\right)$ is $\mathcal{N} = \mathbb{Z}[1, i, j, k]$, and we recall that $\operatorname{disc}(\mathcal{N}/\mathbb{Z}) = [\Lambda : \mathcal{N}]^2 \cdot \operatorname{disc}(\Lambda/\mathbb{Z})$.

Order $\mathcal{O}$	Algebra	$\mathcal{M}(\mathcal{O})$	$\lvert\operatorname{disc}(\mathcal{O}/\mathbb{Z})\rvert$	$[\Lambda : \mathcal{N}]$
$\mathbb{Z}$	$\mathbb{Q}$	$1/4$	1	—
$\mathbb{Z}[i]$	$\mathbb{Q}(i)$	$1/2$	4	—
$\mathbb{Z}[\eta]$ $(\eta = (1+k)/2)$	$\mathbb{Q}(\sqrt{-3})$	$1/3$	3	—
Λ_1 (Hurwitz)	$\left(\frac{-1,-1}{\mathbb{Q}}\right)$	$1/2$	4	2
$\mathcal{N}_1$ (Lipschitz)	$\left(\frac{-1,-1}{\mathbb{Q}}\right)$	1	16	
Λ_2	$\left(\frac{-1,-3}{\mathbb{Q}}\right)$	$2/3$	9	4
$\mathcal{N}_2$	$\left(\frac{-1,-3}{\mathbb{Q}}\right)$	2	144	
Λ_3	$\left(\frac{-2,-5}{\mathbb{Q}}\right)$	$4/5$	—	—

B Output Bounds Comparison

Table 1: Output bounds at $\delta = 1$ for a rank-n $\mathcal{N}_\iota$-lattice, where $V = \operatorname{vol}(\mathcal{L})^{2/n}$.

	$\frac{\lVert\mathbf{b}_i\rVert^2}{\lambda_i(\mathcal{L})^2}$	$\frac{\lVert\mathbf{b}_1\rVert^2}{V^{1/n}}$	$\Omega_{\mathcal{L}}(\mathbf{B})$
	$\mathcal{A}_1 = \left(\frac{-1,-1}{\mathbb{Q}}\right)$		
$\mathbb{Z}$-LLL (rank-$4n$)	$\left(\frac{4}{3}\right)^{4n-1}$	$\left(\frac{4}{3}\right)^{\frac{4n-1}{2}}$	$\left(\frac{4}{3}\right)^{n(4n-1)}$
$\mathbb{Z}[i]$-LLL (rank-$2n$)	2^{2n-1}	$2^{\frac{2n-1}{2}}$	$2^{\frac{n(2n-1)}{2}-1}$
Sublattice-QLLL (rank-n)	2^{n-1}	$2^{\frac{n-1}{2}}$	$2^{\frac{n(n-1)}{4}-2}$
Natural-QLLL (rank-n)	$F_{n+1} \cdot 2^{n-1}$	$F_{n+1}\gamma_{4n} \cdot 2^{n-1}$	$(F_{n+1}F_{n+2})^{n/4} \cdot 2^{\frac{n(n+1)}{4}-2}$
	$\mathcal{A}_2 = \left(\frac{-1,-3}{\mathbb{Q}}\right)$		
$\mathbb{Z}$-LLL (rank-$4n$)	$\left(\frac{4}{3}\right)^{4n-1}$	$\sqrt{3}\left(\frac{4}{3}\right)^{\frac{4n-1}{2}}$	$\left(\frac{4}{3}\right)^{n(4n-1)}$
$\mathbb{Z}[i]$-LLL (rank-$2n$)	2^{2n-1}	$\sqrt{3} \cdot 2^{\frac{2n-1}{2}}$	$2^{\frac{n(2n-1)}{2}-1}$
$\mathbb{Z}[\eta]$-LLL (rank-$2n$)	$\left(\frac{3}{2}\right)^{2n-1}$	$\left(\frac{3}{2}\right)^{\frac{2n-1}{2}}$	$3^{-1/2}\left(\frac{3}{2}\right)^{\frac{n(2n-1)}{2}}$
Sublattice-QLLL (rank-n)	3^{n-1}	$3^{\frac{n-1}{2}}$	$\frac{1}{12} \cdot 3^{\frac{n(n-1)}{4}}$
Natural-QLLL (rank-n)	$F_{n+1} \cdot 3^{n-1}$	$\sqrt{3}\,F_{n+1}\gamma_{4n} \cdot 3^{n-1}$	$\frac{1}{12}(F_{n+1}F_{n+2})^{n/4} \cdot 3^{\frac{n(n+1)}{4}}$

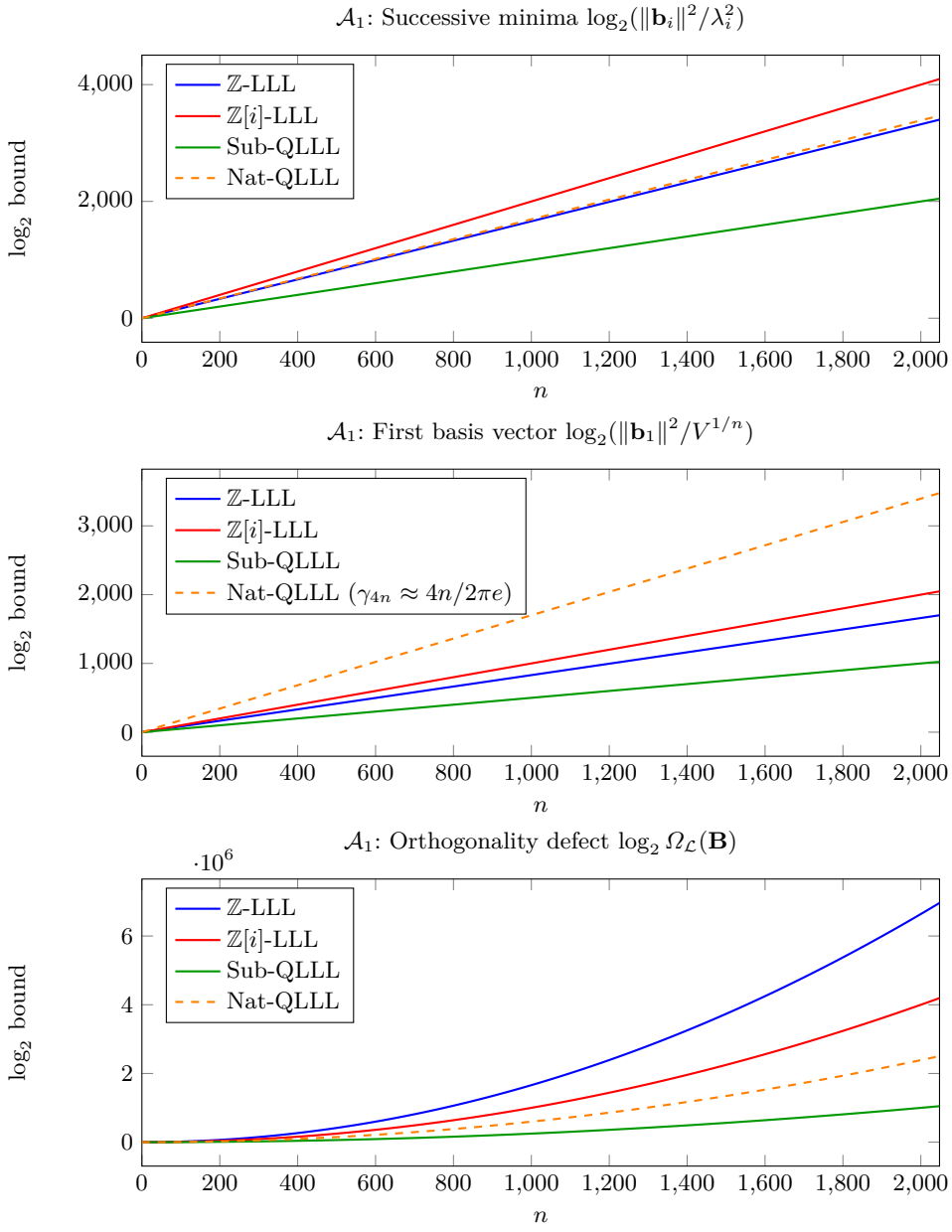


Fig. 1: $\log_2$ of the output bounds from Table 1 for $\mathcal{A}_1 = \left(\frac{-1,-1}{\mathbb{Q}}\right)$. Fibonacci numbers are approximated as $F_k \approx \varphi^k/\sqrt{5}$; the Natural-QLLL first basis vector uses the Minkowski approximation $\gamma_{4n} \approx 4n/(2\pi e)$.

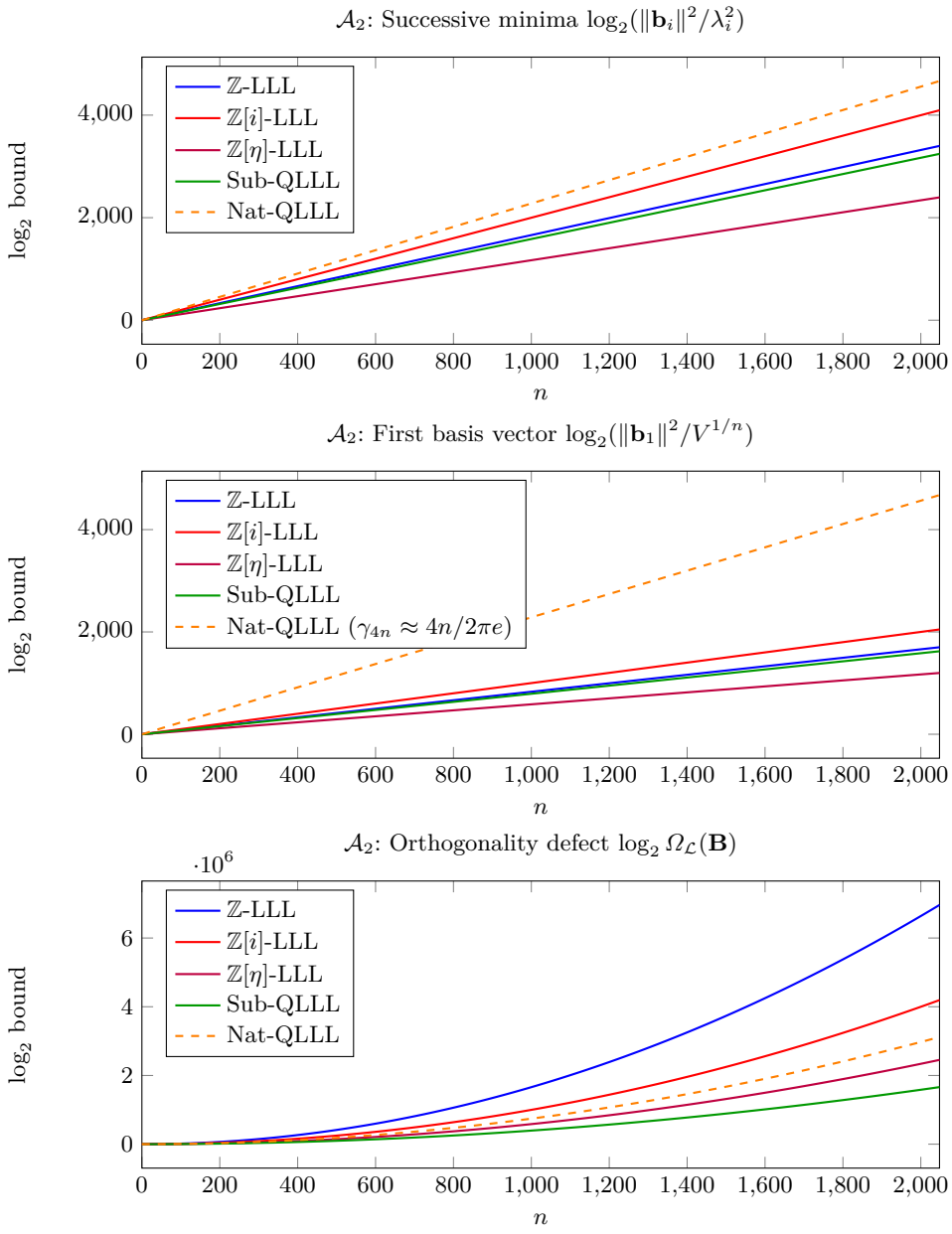


Fig. 2: $\log_2$ of the output bounds from Table 1 for $\mathcal{A}_2 = \left(\frac{-1,-3}{\mathbb{Q}}\right)$. Fibonacci numbers are approximated as $F_k \approx \varphi^k/\sqrt{5}$; the Natural-QLLL first basis vector uses the Minkowski approximation $\gamma_{4n} \approx 4n/(2\pi e)$.

Table 2: QBKZ Hermite factor $\|\mathbf{b}_1\|/\lambda_1(\mathcal{L})$ at matched SVP cost (QBKZ$_\beta$ vs $\mathbb{Z}$-BKZ$_{4\beta}$), using $\gamma_{\beta,\mathcal{N}} \leq (\gamma_{4\beta}/2) \cdot \mathrm{disc}(\mathcal{N}/\mathbb{Z})^{1/4-1/\beta}$.

	$\|\mathbf{b}_1\|/\lambda_1$
$\mathbb{Z}$-BKZ$_{4\beta}$	$\gamma_{4\beta}^{\frac{4n-1}{4\beta-1}}$
QBKZ$_\beta$, Λ_1	$\left(\frac{\gamma_{4\beta}}{\sqrt{2}}\right)^{\frac{n-1}{\beta-1}}$
QBKZ$_\beta$, $\mathcal{N}_1$	$\gamma_{4\beta}^{\frac{n-1}{\beta-1}}$
QBKZ$_\beta$, Λ_2	$\left(\frac{\sqrt{3}\,\gamma_{4\beta}}{2}\right)^{\frac{n-1}{\beta-1}}$
QBKZ$_\beta$, $\mathcal{N}_2$	$\left(\sqrt{3}\,\gamma_{4\beta}\right)^{\frac{n-1}{\beta-1}}$

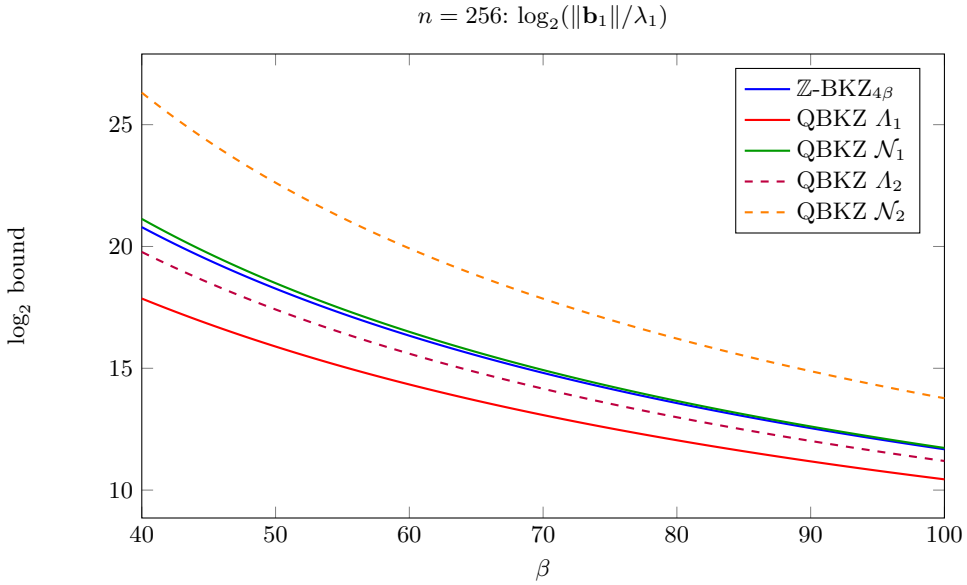


Fig. 3: $\log_2(\|\mathbf{b}_1\|/\lambda_1)$ from Table 2 as a function of β for $n = 256$, using $\gamma_{4\beta} \approx 4\beta/(2\pi e)$.

C Additional Algorithms

Algorithm 8 Swap Vectors

$SwapVectors(\mathbf{R}, \mathbf{U}, k)$:

Swap $\mathbf{u}_{k-1}$ and $\mathbf{u}_k$
Perform Givens' Rotation on $\mathbf{R}_{k-1:k,k-1:k}$ submatrix
Swap $\mathbf{r}_{k-1}$ and $\mathbf{r}_k$, skipping the already updated 2×2 block

Algorithm 9 Modified Quaternion 2-BKZ

$\mathbf{B}_{red} \leftarrow$ 2-$QBKZ(\mathbf{B}, \delta)$) :

$\Omega = 0$
while $\Omega < n$ **do**
  for $i = 1, ..., n-1$ **do**
    $\mathbf{v}_{\Lambda_1} = x_i\pi_i(\mathbf{b}_i) + x_{i+1}\pi_i(\mathbf{b}_{i+1}) \leftarrow GaussReduce(\pi_i(\mathbf{B}_{[i,i+1]}), \Lambda_1)$ ▷ Alg. 5
    $\mathbf{v} \in \mathcal{N}_1 \leftarrow$ Theorem 3 using $(\mathbf{v}_{\Lambda_1})$
    if $\|\mathbf{v}\| < \|\mathbf{b}_i(i)\|$ **then**
      $\mathbf{v}' \leftarrow x_i\mathbf{b}_i + x_{i+1}\mathbf{b}_{i+1}$
      Insert $\mathbf{v}'$ at index i
      Remove linear dependencies
      for $1 \leq j \leq i-1$ **do**
        if $N(\mu_{i,i-j}) > \mathcal{M}(\mathcal{N}_1)$ **then**
          $\mathbf{b}_i \leftarrow \mathbf{b}_i - Q_{\mathcal{N}_1}(\mu_{i,i-j})\mathbf{b}_{i-j}$
        end if
      end for
      $\Omega \leftarrow 0$
    else
      $\Omega \leftarrow \Omega + 1$
    end if
  end for
end while

Algorithm 10 Construct Quaternion Ideal from Rank-2 MLIP Instance [6]

$\mathcal{I}, \mathcal{A} \leftarrow Q_Ideal($
$F = \mathbb{Q}(\zeta_m), K = \mathbb{Q}(\zeta_m + \zeta_m^{-1})$
$G = (\mathbf{G} = \mathbf{B}^*\mathbf{B}, \mathfrak{a}_1, \mathfrak{a}_2)$
$G' = (\mathbf{G}, \mathfrak{b}_1, \mathfrak{b}_2) \cong G$
$)$:

if $3|m$ **then**
  $\mathcal{A} \leftarrow (\frac{-1,-3}{K})$
else if $4|m$ **then**
  $\mathcal{A} \leftarrow (\frac{-1,-1}{K})$
else
  return $\perp$
end if
$\mathcal{O}_\mathcal{A} \leftarrow FindMaximalOrder(\mathcal{O}_F \oplus \mathcal{O}_F j)$ ▷ (Algorithm 4, [6])
$\delta \leftarrow ComputeDet(G, G')$ ▷ (Lemma 2.23, [6])
$\gamma \leftarrow \delta \cdot det(\mathbf{B})$
$q \leftarrow g_{2,2}^{-1}(g_{2,1} - \gamma j)$
$\mathcal{I} \leftarrow \mathfrak{b}_1^{-1}\mathcal{O}_\mathcal{A} \cap q\mathfrak{b}_2^{-1}\mathcal{O}_\mathcal{A}$
return $\mathcal{I}, \mathcal{A}$

Algorithm 11 Quaternion Lattice Basis from a Quaternion Ideal

$\mathbf{B}_\mathcal{I} \leftarrow Q_Ideal_Lattice($
$\mathcal{I} = \mathfrak{b}_1^{-1}\mathcal{O}_\mathcal{A} \cap q\mathfrak{b}_2^{-1}\mathcal{O}_\mathcal{A}$
$)$:

$\mathfrak{c}_1 = \mathfrak{b}_1^{-1}\mathcal{O}_\mathcal{A}$
$\mathfrak{c}_2 = q\mathfrak{b}_2^{-1}\mathcal{O}_\mathcal{A}$
for $i \in \{1, 2\}$ **do**
  $c_i, c_i' \leftarrow$Find two generator representation of $\mathfrak{c}_i$
  $\mathbf{B}_{\mathfrak{c}_i} = HNF(\Sigma^{\mathbb{H}}(c_i, c_i'))\sigma(\omega)$
  $\mathbf{B}_{\mathfrak{c}_i}^\wedge = \mathbf{B}_{\mathfrak{c}_i}(\mathbf{B}_{\mathfrak{c}_i}^H\mathbf{B}_{\mathfrak{c}_i})^{-1}$
end for
$\mathbf{B}_\mathcal{I} = (HNF(\mathbf{B}_{\mathfrak{c}_1}^\wedge | \mathbf{B}_{\mathfrak{c}_2}^\wedge))^\wedge$
return $\mathbf{B}_\mathcal{I}$